

STRyper v. 1.6 User Guide



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License agreement

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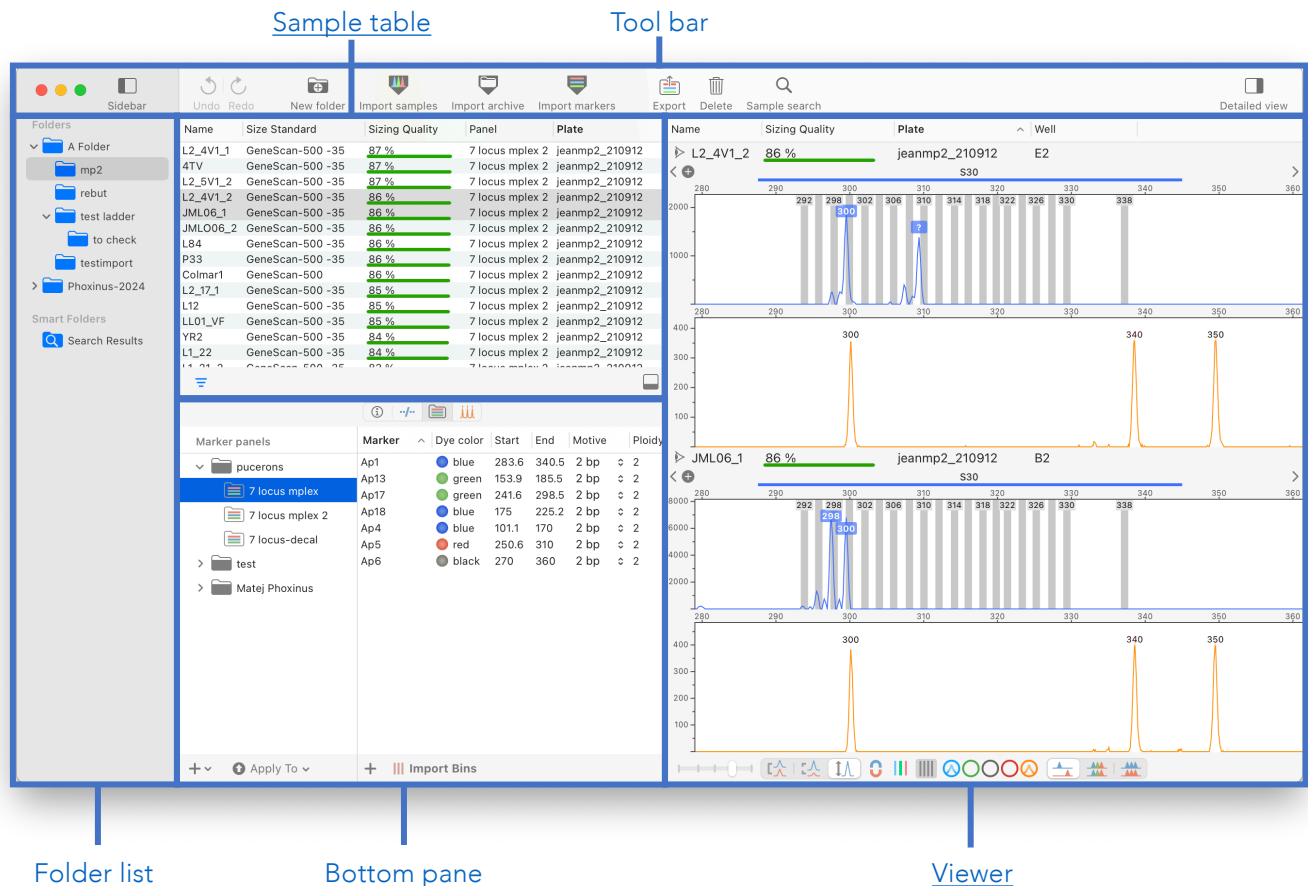
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STRyper's interface

STRyper is designed to manage and analyze chromatograms generated by capillary sequencers for microsatellite genotyping. The application main window is vertically split in three main sections.

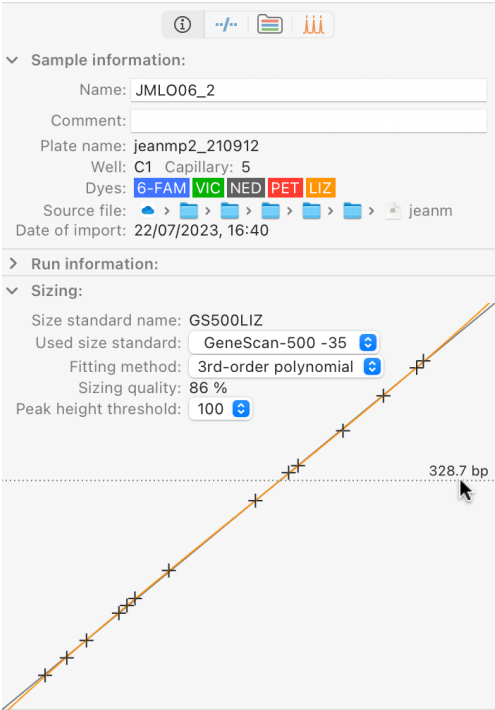


The left sidebar contains the **folder list**.

The mid-section contains:

- [the sample table](#) listing the **samples** (imported chromatogram files) in the selected folder,
- the bottom pane with four tabs:

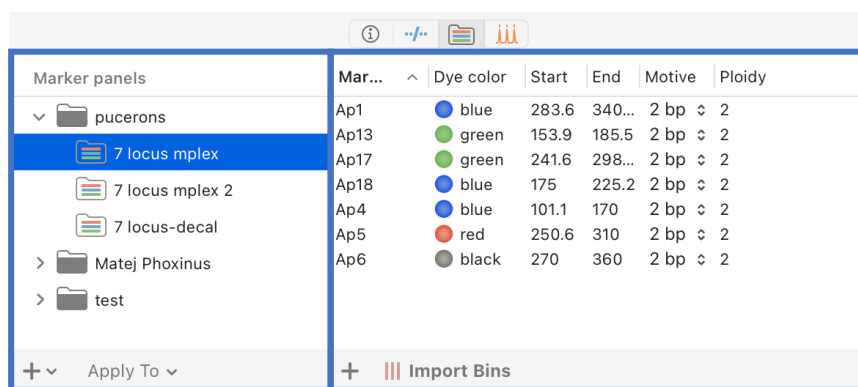
1) [the sample inspector](#) listing information about the selected sample(s),



2) [the genotype table](#), listing genotypes for the sample displayed in the sample table.

Status	Sample	Panel	Marker	Size1	Size2	Allele1	Allele2
●	4TV	7 locus mpl...	Ap2	417.5	431.6	416	430
●	4TV	7 locus mpl...	Ap2	417.4	431.6	416	430
●	Colmar1	7 locus mpl...	Ap2	427.7	427.7	426	426
●	Colmar1	7 locus mpl...	Ap2	428	428	?	?
●	JML06_1	7 locus mpl...	Ap2	417.3	419.3	416	418
●	JML06_1	7 locus mpl...	Ap2	417.4	419.4	416	418
●	JML00...	7 locus mpl...	Ap2	417.4	419.4	416	418
●	JML00...	7 locus mpl...	Ap2	417.4	419.4	416	418
●	L12	7 locus mpl...	Ap2	417.4	427.7	416	426
●	L12	7 locus mpl...	Ap2	417.5	427.7	416	426
⊗	L1_21_2	7 locus mpl...	Ap2				
●	L1_21_2	7 locus mpl...	Ap2	429.6	431.3	428	430
●	L1_22	7 locus mpl...	Ap2	417.5	427.8	416	426
●	L1_22	7 locus mpl...	Ap2	417.4	427.8	416	426
●	L2_17_1	7 locus mpl...	Ap2	417.1	429.4	416	428
●	L2_4V1_1	7 locus mpl...	Ap2	417.3	417.3	416	416
●	L2_4V1_1	7 locus mpl...	Ap2	417.4	417.4	416	416
●	L2_4V1...	7 locus mpl...	Ap2	417.5	417.5	416	416

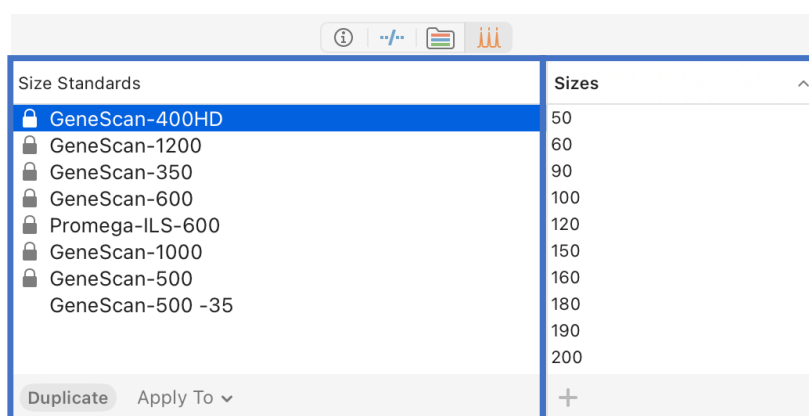
3) [the marker library](#), listing panels of molecular markers, and molecular markers of the selected panel:



[Panel list](#)

[Marker library](#)

4) **the size standard library**, listing available [size standards](#) and the **sizes** of the selected size standard:



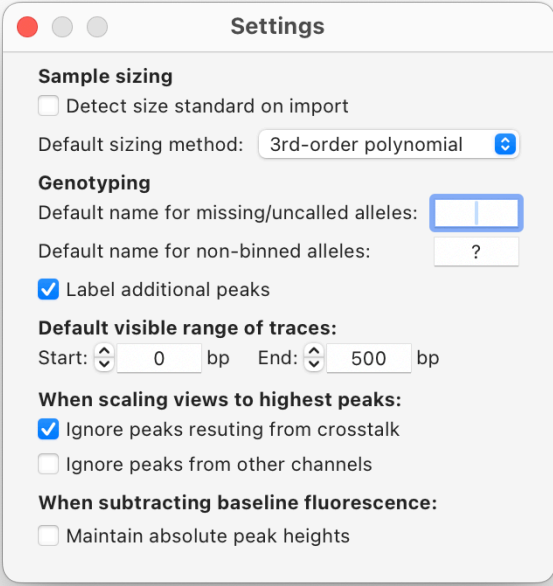
[Size standard library](#)

[Size list](#)

The right pane contains the [viewer](#), which displays the selected samples, genotypes, or molecular markers.

The folder list, the bottom pane and the right pane **can be collapsed/expanded** by pressing the collapse buttons , by dragging their separator, or via the **View** menu in the menu bar.

STRyper also has a **settings window** accessible from the menu bar via STRyper > Settings (⌘,).



The screenshot shows a macOS-style settings window titled "Settings". It contains several sections with various controls:

- Sample sizing**
 - ☐ Detect size standard on import
 - Default sizing method: 3rd-order polynomial (dropdown menu)
- Genotyping**
 - Default name for missing/uncalled alleles: (empty text box)
 - Default name for non-binned alleles: ? (text box)
 - ☒ Label additional peaks
- Default visible range of traces:**
 - Start: 0 bp (spinners)
 - End: 500 bp (spinners)
- When scaling views to highest peaks:**
 - ☒ Ignore peaks resulting from crosstalk
 - ☐ Ignore peaks from other channels
- When subtracting baseline fluorescence:**
 - ☐ Maintain absolute peak heights

Managing chromatograms

Folders

In STRyper, imported chromatograms files are referred to as **samples** and are organized in **folders** . Folders allow you to manage samples by projects however you like. A folder can contain samples analyzed at different sets of molecular markers. To create a folder, do one of the following:

- ▶ choose **File > New Folder** () from the menu bar,
- ▶ place the mouse over the **Folders** section of the [folder list](#) and click the  button 
- ▶ click the **New Folder** button  on the [toolbar](#).

To **add a subfolder to an existing folder** right-click or ctrl-click the folder and choose **Add Subfolder** from the contextual menu.

Folders can be dragged up or down, into, or out of, parent folders.

To **rename a folder**, you may either:

- ▶ click its name,
- ▶ right-click the folder and choose the **Rename** contextual menu,
- ▶ selected the folder and press the enter key,
- ▶ select the folder and choose **Edit > Rename Folder** from the menu bar.

To **delete a folder**, do one of the following:

- ▶ right-click it and choose **Delete Folder** from the contextual menu,
- ▶ select it and choose **Edit > Delete Folder** () from the menu bar,
- ▶ select it and click the trash button  on the [toolbar](#).

Importing samples

STRyper can import chromatogram files ending with the ".fsa", ".FSA", ".hid" or ".HID" extension. These files are imported as **samples** into [folders](#).

Note: HID file support is experimental as the specifications of this format are not public.

To **import samples into a folder**, do one of the following:

- ▶ select the folder and click the **Import Samples** button  on the [toolbar](#), or choose **File > Import Samples...** () from the menu bar,
- ▶ right-click or ctrl-click the folder and choose **Import Samples...** from the contextual menu,

- drag and drop FSA/HID files, or their parent directory, from the Finder into the sample table (if the folder is selected) or onto the folder itself.

You can import chromatogram files by selecting or dropping several directories at once.

Note: If you drag & drop directories from the Finder or select directories from the import window, the chromatograms they contain are imported without creating folders in STRyper. Chromatograms from subdirectories are not imported.

The sample table

Samples of the selected [folder](#) are listed in the [sample table](#). The sample table has a default set of columns showing sample metadata including sample name, plate, well, run date, etc.

Name	Size Standard	Sizing Quality	Panel	Plate
A10				171218
LADDER	GeneScan-500	76 %		
IVC02f	GeneScan-500	84 %	7 locus mplex 2	jeanfinFHIJ_231007
IMO03a	GeneScan-500	85 %	7 locus mplex 2	jeanfinFHIJ_231007
IVC02a	GeneScan-500	85 %	7 locus mplex 2	jeanfinFHIJ_231007
IVC02e	GeneScan-500	85 %	7 locus mplex 2	jeanfinFHIJ_231007
IVC02b	GeneScan-500	85 %	7 locus mplex 2	jeanfinFHIJ_231007
IMO03d	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
ILaP01c	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
IMO03e	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
IVC02d	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
IOR03	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
ILaP01b	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
IMO03c	GeneScan-500	86 %	7 locus mplex 2	jeanfinFHIJ_231007
IOS05h	GeneScan-500	87 %	7 locus mplex 2	jeanfinFHIJ_231007
ILaP01a	GeneScan-500	87 %	7 locus mplex 2	jeanfinFHIJ_231007

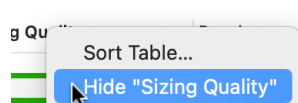
The signification of each column is explained by a tooltip that appears when the column header is hovered. You may refer to the sequencing application manual for a more detailed explanation on certain columns such as "Protocol", "Result group", "Owner", etc.

To copy the textual content of the table, select rows, then Edit > Copy (⌘C) from the menu bar. The copied content can be pasted to a text editor or a spreadsheet application.

To reorder columns, drag them with the mouse.

The resize columns to fit their content, right-click or ctrl-click the table header and choose Size Columns to Fit Content from the contextual menu.

To show/hide columns, right-click or ctrl-click the table header. This pops up a contextual menu allowing to hide the clicked column or to check/uncheck the columns you want to show/hide.



To sort the table by a column, click its header.

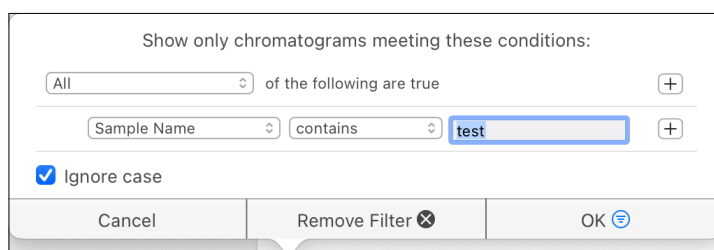
To sort the table by several columns, right-click or ctrl-click the table header and choose **Sort Table...** from the contextual menu. This spawns a popover that lets you change the priority of columns by dragging rows up or down.



You may change the sorting order (ascending descending) by clicking the button with a chevron pointing down or up, and add remove sorting criteria using the + and – buttons: .

Filtering the sample table

To display samples meeting certain criteria, press the “filter” button  below the [sample table](#). A popover lets you to define filter criteria, which can be added by clicking a + button.



A filter on samples is associated to a specific folder. Upon selecting a folder, the filter associated to it (if any) applies. If a folder has a filter applied to it, the filter button changes appearance: .

Note: a filter on samples is not dynamic. If sample metadata are modified so that they meet or no longer meet filter criteria, you have to reapply the filter by clicking the filter button, or select another folder and go back to the previous selection, to update the sample table.

To remove a filter applied to samples, click the filter button  and then the **Remove Filter** button on the popover.

Moving, deleting and copying samples

To move samples between folders, select samples and drag them onto a folder in the [folder list](#).

To delete samples, select them in the sample table and either:

- ▶ click the trash button  on the [toolbar](#),
- ▶ chose **File > Delete Sample(s)** () from the menu bar,
- ▶ right-click or ctrl-click the samples and selected **Delete** from the contextual menu.

Note: any deletion in STRyper can be undone, but there is no trash folder that you can browse to retrieve deleted items.

You may **copy & paste samples** to duplicate them. To do so, select samples from the [sample table](#) and choose Edit > Copy (⌘C) from the menu bar, or right-click samples and choose Copy from the contextual menu. You can then **paste copied samples** via Edit > Paste (⌘V).

Note: Upon pasting, the copied samples are duplicated in their current state, not in the state they had when you copied them. The sample's [genotypes](#) are copied as well.

Sample search and smart folders

To find **samples according to different criteria** across the whole database, press the search button  on the [toolbar](#) or choose File > Find Samples... (⌘F) from the menu bar. This spawns a sheet allowing you to define the same criteria as those of the [filter popover](#).

Search results appear in a special folder called a **Smart Folder** . **A new smart folder is created upon every search.**

Smart folders in STRyper work like smart folders in the Finder, which are created after a spotlight search is saved. This means that **samples meeting the search criteria are automatically listed in the smart folder after it is created.** For certain search criteria however (parent folder name, size standard, panel name), the smart folder contents may not update immediately and only update once the application saves the database (which occurs at most every 30 seconds).

You may also initiate a search by **adding a new smart folder**. To do so, hover the Smart Folders section in the folder list and click the  button: 

A smart folder can be renamed and moved just like a regular folder. However, you cannot drop any item (folder, samples) into a smart folder neither can you move a smart folder into another folder.

A smart folder is preserved between app launches, which means you have to [delete a smart folder](#) to **discard search results**. However, you need not create a new smart folder every time you want to find samples.

To **modify the search criteria of an existing smart folder**, do one of the following:

- ▶ select the smart folder and click the  button at the below the sample table,
- ▶ right-click the smart folder and choose Modify Search Criteria from the contextual menu,
- ▶ select the smart folder and choose Edit > Edit Smart Folder... from the menu bar.

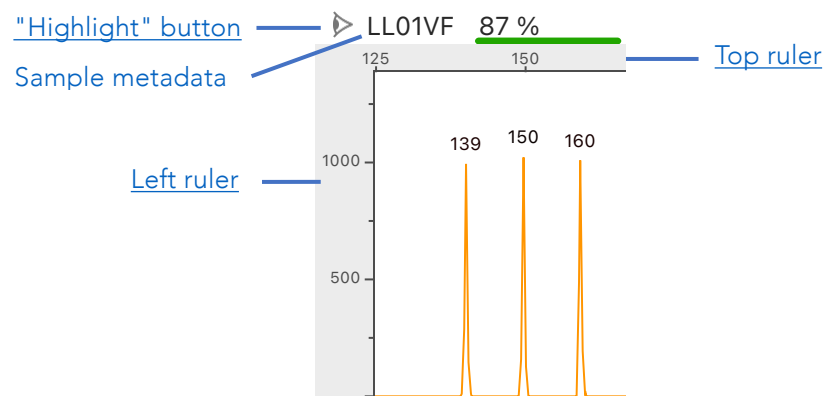
Important: a smart folder does *not* contain copies of samples present in other folders. It just gathers samples meeting the search criteria, which means that **deleting samples from a smart folder removes them from the database**. However, deleting a smart folder itself does not delete any sample.

To reveal samples from a smart folder in their original folder, right-click or ctrl-click the sample(s) and choose Reveal in Parent Folder from the contextual menu. This item is disabled if clicked samples belong to several folders.

Visualizing chromatograms

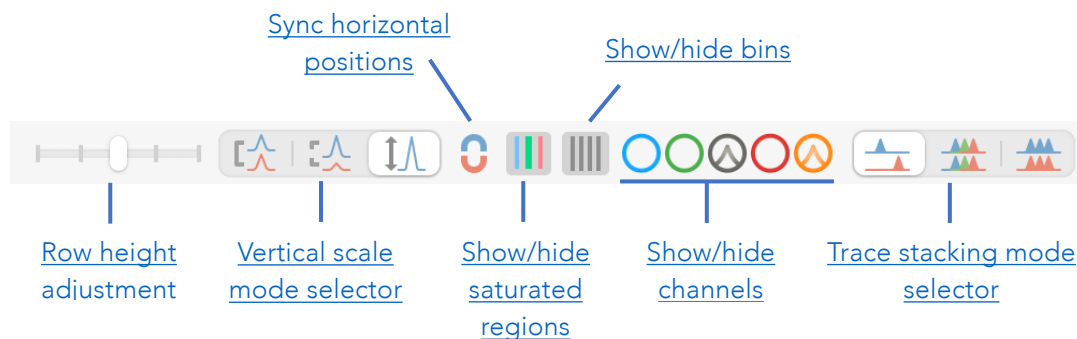
The viewer

Chromatograms of samples that are selected in [the sample table](#) are displayed in the **viewer**, which occupies the [right pane](#). The viewer shows **traces** (fluorescence curves).



If samples are not [stacked](#), the viewer also shows sample metadata like the [sample table](#) does (with the same options for [hiding columns](#)).

When samples are displayed, the bottom of the right pane presents these buttons:



The viewer can show the content from several tables: the sample table, the [genotype table](#) and the [marker library](#). The current source of content is indicated by a vertical line at the table's right edge.

Plate	Well
CD_LIZ_Plaque3_23-01-25	A2
CD_LIZ_Plaque3_23-01-25	E1
CD_LIZ_Plaque3_23-01-25	E9

Adjusting row height

The viewer can show up to five rows with traces at once. The number of rows is set via the row [height adjustment slider](#). Row height is constrained between 40 and 1000 pixels.

Showing/hiding channels

The viewer can display chromatograms containing fluorescent data from four or five **channels** (colors, or emission wavelengths of fluorescent dyes).

To display or hide a channel, tick/untick the button with the corresponding color: 

To display a channel and hide the others at the same time, click the colored button with the option key (⌘) pressed.

Note: you cannot hide all channels.

Stacking curves in the same row

To display each trace in a separate row (the default mode), press the left segment  of [the stack trace mode selector](#) or choose Trace > Stack Traces > One Trace per Row (⇧⌘T) from the menu bar. This mode allows visualizing amplicons and [molecular ladder](#) fragments individually.

To display all channels from each sample in a single row, press the middle segment  of the selector or choose Trace > Stack Traces > One Sample per Row (⇧⌘S) from the menu bar. This mode can be useful to compare samples with respect to the peaks they present at various channels.

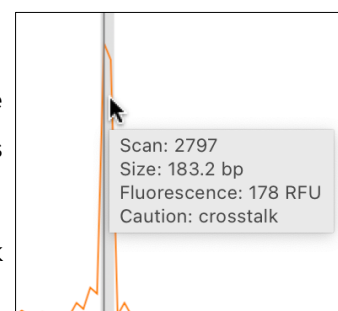
To display traces of the same channel from several samples in the same row, press the right segment  of the selector or choose Trace > Stack Traces > One Channel per Row (⇧⌘C) from the menu bar. In this mode, sample metadata is not displayed since each row can show several samples (unless a single sample is displayed), and the header of the [viewer](#) is replaced by a text indicating the number of samples stacked. This mode can help to identify all peaks induced by amplicons of the same molecular marker and to [define bins](#), among other uses.

Note: the viewer does not display more than 400 traces per row.

Showing peak information

STRyper detects peaks in traces. If the viewer shows a [single trace per row](#), hovering a peak with the mouse spawns a tooltip listing basic information about the peak:

- “Scan” is the number of data points that were recorded by the sequencer up to the peak tip (represented by a vertical line). It represents the time since the start of the fluorescence recording.
- “Size” is the estimated size of the DNA fragment that caused the peak and is inferred from the [molecular ladder](#).



- “Fluorescence” is the fluorescence intensity at the peak tip in Relative Fluorescence Units (RFU).
- “Caution: crosstalk” indicates that the peak likely results from [crosstalk](#), i.e., it does not represent a DNA fragment from the channel that is displayed, but from another channel.

To show/hide peak information, select/deselect **Traces > Peak Information Tooltips** from the menu bar.

Adding peaks manually

STRyper does not detect peaks whose fluorescence level is less than 100 RFU. Adding a peak manually could be necessary if you consider that it represents an amplicon or a molecular ladder fragment.

To force the detection of a peak in a region where no peak is detected, you can:

- ▶ right-click below the tip of a peak and choose **Add Peak here** from the contextual menu.
- ▶ If you use a trackpad with “force click” enabled, perform a force click (deep press) below the tip of the peak.

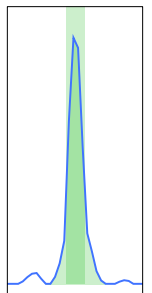
Either action adds the peak at the clicked location. If the clicked location does not appear to correspond to a peak, or if there is already a peak at the clicked point, the action has no effect.

Showing saturated regions

Intense fluorescence may **saturate the sequencer’s camera**. In this case, the peak that saturated the camera may be clipped. Its height measured in RFUs therefore underestimates the quantity of DNA and the location of its tip (which is used to compute the size of the corresponding fragment) is inaccurate. Saturation may also induce **crosstalk between channels**, hence peaks that do not represent DNA fragments. For these reasons, it is important to know where saturation occurred.

To show/hide **regions where the fluorescence has saturated the sequencer’s camera**, press the [Show/hide saturated regions button](#) .

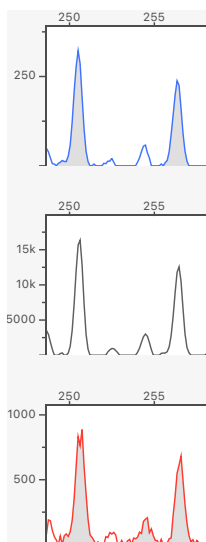
Saturated regions are represented by vertical rectangles whose color corresponds to the channel that likely saturated the camera. Saturated regions are not displayed if several chromatograms are [stacked on the row](#).



Tip: to avoid saturation, increase the dilution of amplicons.

Showing peaks resulting from crosstalk

Peaks in traces may not represent DNA fragments but **crosstalk between channels**. STRyper fills the area under such peaks with the color of the channel that likely induced crosstalk. It is the case for the black channel on the screen capture below. The channel that caused crosstalk is inferred from the intensity and location of peaks from different channels.



Peaks induced by crosstalk are ignored by the application during [genotype calling](#) or determination of fragments in the DNA ladder.

To enable/disable the outlining of peaks resulting from crosstalk, activate/deactivate **Traces > Outline Crosstalk Peaks** from the menu bar. This display setting does not change how peaks are treated internally.

Note: STRyper might occasionally consider legit peaks as resulting from crosstalk in case DNA fragments labelled with different dyes generated peaks at the same locations (in base pairs). Conversely, certain peaks resulting from crosstalk may not be detected as such.

Subtracting baseline fluorescence level

In chromatograms, the fluorescence level between peaks may not always be zero, and represents some **baseline**. By default, STRyper automatically subtract the baseline fluorescence level to make peaks stand out more. In this case, the curves displayed in the [viewer](#) do not represent raw fluorescence data.

To disable baseline subtraction, deactivate **Traces > Subtract Baseline Fluorescence (^R)** from the menu bar.

Note: crosstalk between channels and other types of interference can induce negative fluorescence values. STRyper draws negative values at zero RFU regardless of the setting to subtract the baseline fluorescence level.

By default, the height of peaks is not affected by baseline fluorescence level subtraction, which therefore increases peak heights relative to the new baseline. If you want to maintain relative peak heights rather than absolute peak heights, uncheck the **Maintain absolute peak heights** checkbox in the [application settings](#).

Adjusting the vertical scale

The [left ruler](#) indicates the fluorescence intensity in RFUs.

To change the vertical scale hence change the height of curves, click and drag vertically on the left ruler.

To adjust the scale automatically such that the tip of the highest visible peak appears near the top of the row, double-click the left ruler. This action has no effect if no peak is detected in the visible part of the trace(s).

There are three modes to adjust the vertical scale of rows in the viewer:

To make all rows use the same vertical scale, press the left-hand button  of the [vertical scale mode selector](#) or choose **Traces > Vertical Scales > Synchronized** (⌘S) from the menu bar. Scales are synchronized only after you adjust the vertical scale of a row.

To adjust the vertical independently for each row, press the middle button  of the selector or choose **Traces > Vertical Scales > Independent** (⌘I) from the menu bar. Subsequent adjustments you make to the vertical scale affect the corresponding row only.

To automatically adjust the vertical to the highest visible peak, press the right-hand button  of the selector or choose **Traces > Vertical Scales > Scale to Highest Peaks** (⌘P) from the menu bar. In this mode, you can still adjust the vertical scale by clicking and dragging the left ruler, but the row rescales automatically upon scrolling. By default, this mode ignores peaks resulting from [crosstalk](#) and peaks in other channels when [visualising genotypes](#). This can be changed in the application [settings](#).

Horizontal positioning and zooming

For samples that have been [sized with a size standard](#), the [top ruler](#) shows graduations in base pairs to indicate the size of DNA fragments that induced peaks in traces. The ruler also shows the size corresponding to the current mouse position.

To zoom in/out horizontally, do one of the following:

- ▶ Place the mouse pointer over the traces and either:
 - pinch with two fingers, if this gesture is enabled in the macOS trackpad settings.

- double-tap with two fingers if this gesture is enabled in the trackpad settings. Hold the option key (⌥) while tapping to zoom out.
 - scroll vertically with the option key (⌥) pressed.
- ▶ click the [top ruler](#) where the cursor is a magnifying glass. Press the option key while clicking the top ruler to zoom out.
 - ▶ cover the area you wish to zoom to by clicking and dragging the mouse on the top ruler.



The default visible range of traces (by default, 0 to 550 base pairs) can be specified in the application [settings](#). This range must be at least 2 base-pair wide and is constrained to 0 – 1500 base pairs.

To zoom to the default range you can either:

- ▶ double-click the [top ruler](#) with the option key (⌥) pressed.
- ▶ Right-click or ctrl-click the top ruler and choose **Zoom to Default Range** from the contextual menu.

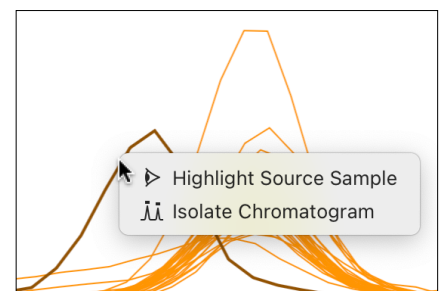
To synchronize the horizontal scale and position across rows, press the [Sync horizontal positions button](#) such that it shows this symbol:  or choose **Traces > Synchronized Horizontally** from the menu bar. This setting has an effect only after you change the scale/position on a row.

Highlighting/isolating a sample

Sometimes, a sample displayed in the [viewer](#) may not be visible in the [sample table](#) if this table or the viewer has been scrolled. In addition, the viewer may show traces from different samples on the same row, and you may want to know which sample a trace corresponds to.

To highlight a sample displayed in the viewer in the sample table:

- ▶ click the [Highlight button](#)  at the top left of the traces.
- ▶ If the traces from several samples are [stacked on the same row](#), right-click or ctrl-click a curve and choose **Highlight Source Sample** from the menu. You may alternatively perform a “deep press” on the curve on a Mac equipped with a force touch trackpad.

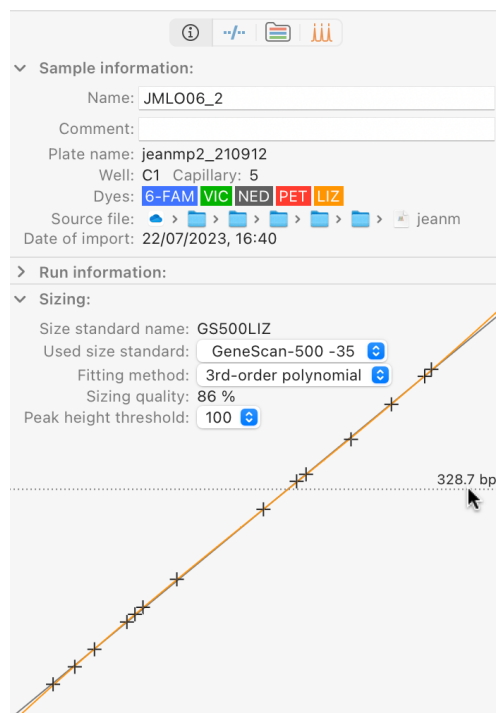


These actions highlight, with a white frame, the sample in the sample table.

To isolate a trace when several samples are stacked on the same row, right-click or ctrl-click a curve and choose **Isolate Chromatogram** from the menu. This action deselects the other samples in the sample table, such that only this sample appears in the viewer.

The sample inspector

The **sample inspector** displays information about selected samples. This information is organized in three sections: Sample information, Run information and Sizing.

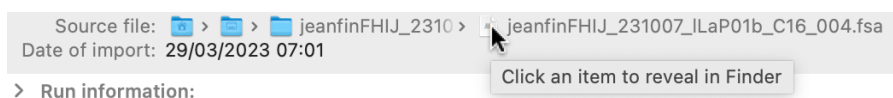


To show the sample inspector, click the left button ⓘ of the segmented control of the bottom pane or choose View > Sample Inspector (⌘⇧A) from the menu bar.

To collapse or expand a section, click the triangle next to the section title.

To reorder sections vertically, drag their header with the mouse.

Tip: to reveal the source file of a selected sample in the Finder, click the file icon in the Sample information section at the Source file field.



Sample sizing

To compute fragment sizes in base pairs, a chromatogram must contain fluorescence data from a **molecular ladder** added to amplicons. The molecular ladder is labelled with a fluorescent dye corresponding to the last (red or orange) channel of the chromatogram. The sizes of fragments of the molecular ladder are defined in a **size standard**. Available size standards are listed in the [size standard library](#).

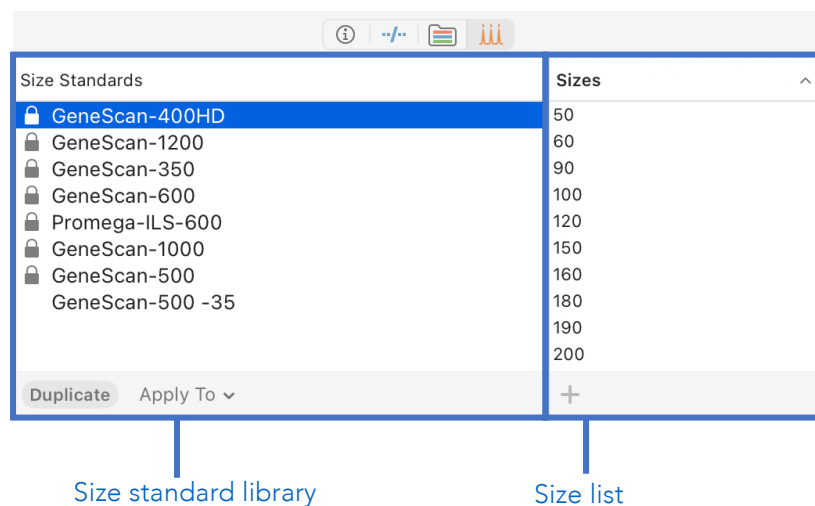
Depending on how sample metadata were specified in the sequencing application before the run, the type of molecular ladder added to a given sample is encoded in the FSA file via the name of the size standard (e.g., "Genescan-500 LIZ"). When the chromatogram is imported, STRyper can use this name to identify molecular ladder fragments given this size standard, if the "Detect size standard on import" option is enabled in the application [settings](#). If you enable this setting, make sure to specify the correct size standard name in the sequencing application before your runs. Otherwise, STRyper may use inappropriate size standards for sizing.

The name of the size standard encoded in a chromatogram file appears in the **Sizing** section of the [sample inspector](#) at the **Size standard name** field. The **Size standard** column of the [sample table](#) shows the name of the sized standard that is used for sizing, which is missing if no size standard has been applied (see below).

Applying a size standard to samples

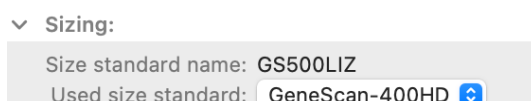
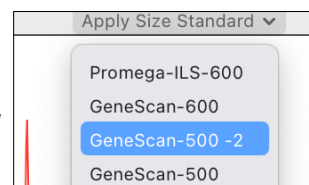
If no size standard was specified or detected, an imported chromatogram is not "sized" (the sizes of fragments are not computed) and the sample cannot yet be genotyped. The **Size standard** column of the sample table shows no information and sizes in base pairs do not appear in the top ruler above traces. You have to **apply a size standard**. Applying a size standard assigns molecular ladder fragments (peaks in the trace) to sizes listed in the standard.

Available size standards are listed in the **size standard library**, which you access by clicking the right-hand button  of segmented control of the bottom pane or by selecting **View > Size Standard Library** (⌘⌘L) in the menu bar. The sizes composing the selected standard appear in the **size list**.



To apply a size standard to samples, do one of the following:

- ▶ From the size standard library, drag and drop the row corresponding to a size standard onto the [sample table](#). This applies the size standard to all samples from the table.
- ▶ Select a size standard from the list, click the **Apply To** button below the list and select the target samples from the contextual menu (either all samples listed in the table or only selected samples).
- ▶ Right-click samples from the sample table and chose the size standard from the **Apply Size Standard** drop-down menu. This applies the size standard to all clicked/selected samples.
- ▶ Select samples from the sample table and select a size standard from the **Apply Size Standard** button that is visible above traces in the [viewer](#). This button is not visible if at least one of the samples displayed at the row already has a size standard.
- ▶ Go to the [sample inspector](#) at the Sizing section and select an item from the **Used size standard** popup button. This applies the size standard to all selected samples:



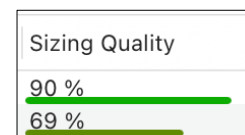
Note: applying a size standard reassigns the DNA ladder fragments of samples that were already sized (with the same size standard or another).

Sizing quality

Applying a size standard does not guarantee that a sample is sized properly. Some errors may occur:

- The size standard does not correspond to the molecular ladder used.
- The molecular ladder was used in too low quantity for all peaks to be detected.
- Problems during electrophoresis.
- Errors in attribution of peaks to size standard fragments by the application.

To indicate possible errors, a score of **sizing quality** from 0 to 100% appears in the Sizing Quality column of the [sample table](#) and the [viewer](#). Sizing quality is



represented by a gauge whose color varies from red to green depending on the score.

This quality score evaluates how much the size attributed to a any molecular ladder fragment is consistent with the fragment's [scan](#). The score also considers the number of sizes that could be attributed to fragments. See, [the original publication of STRyper](#) for more details.

Important: size attributions to molecular ladder fragments **must** be [checked visually](#) regardless of the score. High scores do not guarantee the absence of errors.

Tip: sizing quality can also be evaluated visually by checking the [fitting curve](#).

Sample sizing failure

Application of a size standard might not result in the [top ruler](#) showing graduations in base pairs. This means that **sizing has failed** or that the **sizing quality score is too low** (0%). Sample sizing fails if not enough peaks of the molecular ladder could be assigned to sizes of the size standard. Sizing quality of 0% generally means that at least one size (in base pairs) is assigned to a peak this is clearly incorrect. These problems are indicated by:

- no value or 0% for the sizing quality score,
- "Sizing failed" indicated in the Sizing section of [the sample inspector](#), instead of the fitting curve,
- "Sizing failed" or "Sizing too poor" indicated on the top ruler instead of graduations.

Genotypes cannot be [called](#) on a sample if sizing failed.

If you suspect that sizing failed because the fluorescence signal is too weak, you may lower the **peak height threshold** in the Sizing section of the sample inspector. This threshold corresponds to the minimal fluorescence level a peak must have to be detected. The default value is 100. Changing this setting affects all selected samples and trigger the detection of peaks of the molecular ladder for these samples.

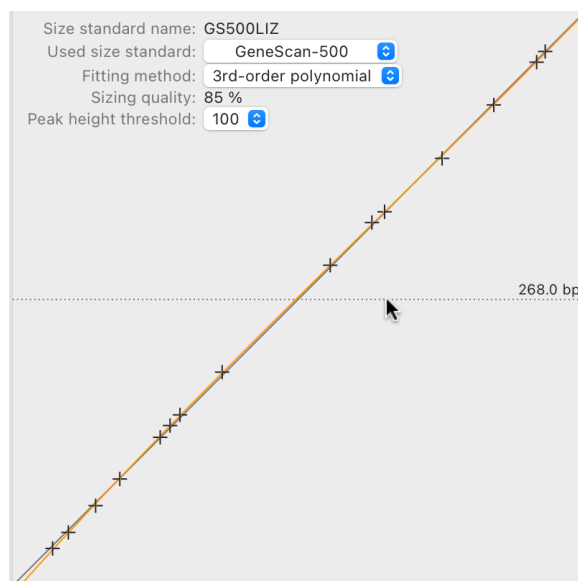
Peak height threshold: 100 

In case of sizing failure, inspect the trace of the molecular ladder in the [viewer](#). A failure is often associated with electrophoretic problems that prevent the detection of molecular ladder fragments (peaks may be missing or have irregular shapes). In such cases, amplicons should be submitted to a another sequencer run.

Sizing failures may be fixed by [manually assigning](#) molecular ladder fragments to sizes.

The fitting curve

STRyper draws a **fitting curve** to show how much the migration speed of molecular ladder fragments is consistent with their theoretical size. To display the fitting curve of a selected sample, go to the [sample inspector](#) and expand the Sizing section.



In the above image, each cross represents a molecular ladder fragment that was assigned a size of the size standard. Its horizontal position is the [scan number](#) at the tip of the peak caused by the fragment. Its vertical position is the fragment's theoretical size in base pairs. Using these points, a [fitting method](#) estimates the size (in base pairs, Y axis) for every scan of the chromatogram. This relationship is represented by the fitting curve, whose color corresponds to the dye of the molecular ladder. The dark grey line represents the relationship inferred via [linear regression](#) (if linear regression is chosen as a fitting method, this line is not drawn).

The curve should pass as close as possible to every point. Points departing noticeably from the curve may indicate that a size was not attributed to the correct fragment of the ladder. In this case, size assignments to DNA fragments should be [corrected manually](#).

Choosing the fitting method

The assignments of molecular ladder fragments to sizes via application of a size standard allows deducing the sizes of other fragments, namely amplicons. This is achieved by a **fitting method**, which computes a size corresponding to each scan (fluorescence data point) of a chromatogram. Three fitting methods are available:

- **The linear regression** (1st order polynomial) method assumes a linear relationship between scan number (n) and size (s): $s = a + bn$.

- **The 2nd order polynomial** method assumes a relationship of the form: $s = a + bn + cn^2$.
- **The 3rd order polynomial** method (default) assumes a relationship of the form: $s = a + bn + cn^2 + dn^3$.

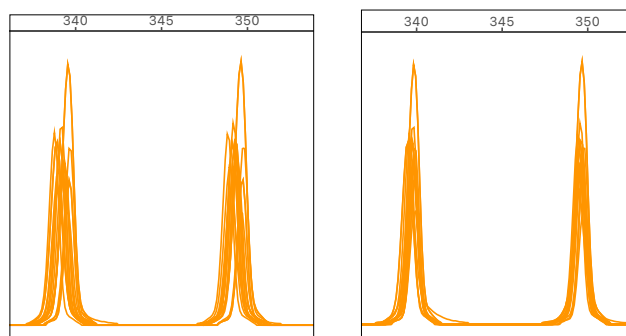
To apply a fitting method on samples, do one of the following:

- ▶ Right-click samples from the [sample table](#) and choose a fitting method in the Fitting Method contextual menu.
- ▶ Select samples from the sample table, go to the [sample inspector](#), Sizing section and choose a value from the Fitting method popup button. Fitting method: 3rd-order polynomial ▾

Applying a fitting method triggers the re-computation of sizes. **The default fitting method** can be defined in the application [settings](#).

The fitting method to choose depends on the consistency between estimated sizes of equivalent fragments from different samples.

Tip: to evaluate consistency of a fitting method, you can [stack all samples](#) from the same run/plate in the same row and [show the channel](#) of the molecular ladder. Then, try different fitting methods and select the one for which peaks superimpose the most, as it is the case for the 3rd order polynomial on the right-hand image below. The lineage regression (left-hand image) estimates sizes that are less consistent.



When in doubt, use the 3rd order polynomial, as this method better considers variations and anomalies in the electrophoresis efficiency.

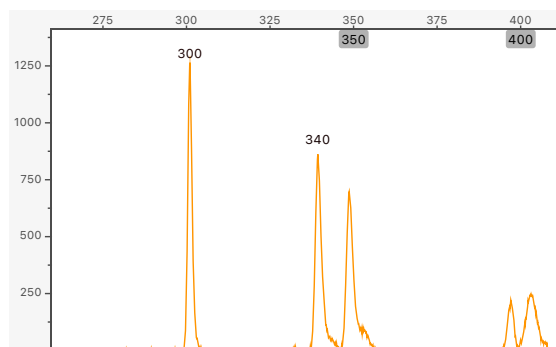
Important: use the same fitting method for all samples analyzed at the same molecular marker, regardless of the run. This is because different fitting methods can yield sizes differing by several base pairs for the same allele.

Checking and assigning sizes in molecular ladders

You **must** check for errors in the assignment of peaks to sizes in the molecular ladder for each sample, regardless of the [sizing quality score](#).

Tip: samples from the same run/plate and using the same molecular ladder should show very similar sizing scores. Scores that are noticeably lower than the majority likely indicate sizing errors or electrophoresis problems.

To visualize sizes attributed to ladder fragments, select samples and [display the channel](#) corresponding to the molecular ladder (red or orange). Make sure to display [a single trace per row](#). The size attributed to each fragment appears as a label above the corresponding peak. Sizes that were not assigned to fragments (because no suitable peaks were found) appear in grey under the top ruler.



To assign a size to a peak, grab the corresponding size label and drop it onto the peak. If the peak already has a size assigned to it, this size is replaced by the new size and becomes unassigned.

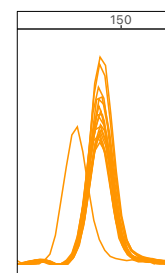
Note: STRyper enforces increasing sizes from left to right, e.g., you cannot assign size 200 to a peak that is at the right of a another that has size 250. You first have to remove size 250 from the other peak.

To remove a size assigned to a fragment, do one of the following:

- ▶ select the corresponding size label and choose **Edit > Remove Ladder Size** (⌘) from the menu bar, or click the trash icon  on the [toolbar](#),
- ▶ double click the corresponding size label or the peak underneath,
- ▶ drag the corresponding size label above the row showing traces.
- ▶ Right-click the size label or the peak underneath and choose **Remove Size** from the contextual menu.

After each change in size assignment, the sizing is automatically recomputed.

Tip: to quickly check for errors in size assignments, [stack all samples in the same row](#) and check whether peaks of the ladder superimpose neatly. Certain peaks appearing off, as in the example on the right, may indicate errors in peak assignments for the corresponding sample(s). You may [isolate samples](#) corresponding to these peaks.



Defining custom size standards

STRyper includes the specifications of several commercial molecular ladders. These size standards cannot be modified nor deleted. If you use molecular ladders that do not conform to these size standards, you need to **define your own size standards**.

You may also define **size standard variants** that omit certain sizes. For instance, the GeneScan-500 molecular ladder includes a 250-bp fragment, yet this size is not specified in the size standard, as

[recommended by the vendor](#). Short fragments (<50 bp) may also be difficult to assign with confidence, due to noise at the start of chromatograms. If the alleles you analyze are much longer (>100 bp), you may for instance use a GeneScan-500 variant that omits size 35 bp.

To define your own size standard, duplicate an existing one with the **Duplicate** button below the [size standard library](#). You may alternatively select a size standard and choose **Edit > Copy** (⌘C) from the menu bar, or right-click the size standard and select **Copy** from the contextual menu. Then choose **Edit > Paste** (⌘V) from the menu bar to paste a copy of the size standard.

On this copy, sizes can be added, removed, or changed in the [size table](#).

To remove sizes from a size standard, do one of the following:

- ▶ select the size(s) and choose **Edit > Delete Size** (⌘⌫) or click the trash icon  on the [toolbar](#).
- ▶ Right-click the size(s) and choose **Delete** from the contextual menu.

To add a size to the standard, select the size standard and click the **+** button below the size table.

To modify a size, select its cell, click on the number and enter a new value. You may alternatively right-click the size and choose **Edit Size** from the contextual menu.

139
150
160
200

Note: a size of a size standard must be at least 10 bp and cannot exceed 1500 bp. A size standard must have at least four sizes and cannot have duplicate sizes.

You can **delete a custom size standard** by selecting it and clicking the trash button on the toolbar, choosing **Edit > Delete Size Standard** (⌘⌫) from the menu bar, or right-clicking it and choosing **Delete** from the contextual menu.

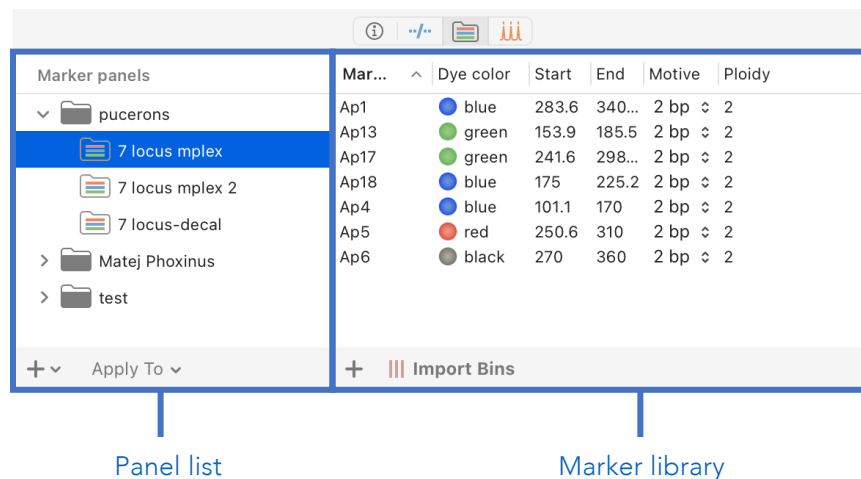
Note: deleting a size standard does not affect the sizing of samples to which it was applied, but the action removes information from the **Size Standard** column of the sample table.

Microsatellite markers

Microsatellite markers and panels

For genotyping, the **microsatellite markers** that you analyze must be specified in STRyper. As a chromatogram often contains data obtained from the amplification of several markers, markers are grouped in **panels**. A panel is a multiplex of markers amplified together or pooled post-PCR. Even a marker amplified alone (not in multiplex) must belong to a panel, which therefore contains a single marker.

You can access **the panel list** by clicking the third button on the segmented control:  or by choosing View > Marker Library (⌘⌘M) from the menu bar.

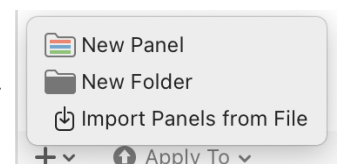


Creating marker panels

A **panel** is a special folder containing molecular markers. If STRyper does not find any panel in the database, it creates one called "New Panel". This panel contains no maker, but markers can be [added to it](#). Like samples, panels can be organized in folders.

To create a new panel or folder, click the **+** button below the [panel list](#).

Panels and folders can be renamed, deleted and moved like [sample folders](#).



To add a panel to an existing folder, right-click the folder and choose Add Panel from the contextual menu.

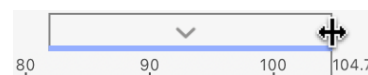
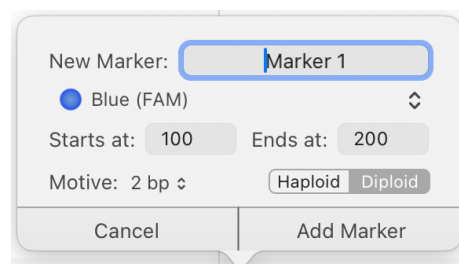
Tip: you can also [import panels](#) exported from STRyper or GeneMapper.

Defining molecular markers

In STRyper, **molecular markers** must be defined so that the application knows where to look for alleles in chromatograms. A marker specifies the **channel** (color) of the **fluorescent dye** that labels the primers used to amplify a microsatellite marker, and the **range** (in base pairs) where alleles are expected, the length of the **repeat motive**, as well as the **ploidy** of the locus, which defines the number of alleles per individual. A marker also has a name.

To add a marker to a panel, do one of the following:

- ▶ select the panel from the [panel list](#) and click the **+** button below [the marker library](#). A popover lets you define the name, the color (dye), the range, the length of the repeat motive and the ploidy of the new marker.
- ▶ In the [viewer](#) showing the appropriate channel of a sample that has the panel [applied to it](#), click the **+** button at the top left of the trace. Then, click and drag on the area where the mouse cursor shows a green '+' sign to define the range of the new marker. Once you release the mouse button, a popover lets you define attributes of the marker, except its color.



Note: The ploidy and color of a marker cannot be changed after its creation.

Important:

- Specifying the correct length of the **repeat motive** is necessary for [genotype calling](#).
- If you expand a multiplex with new markers after having already genotyped samples, you should [create a new panel](#) (for instance, by copying the original panel, see below) rather than adding markers to the existing panel. This is because adding a marker to a panel automatically creates a [genotype](#) at this marker to each sample [associated with the panel](#).

To **delete a marker**, select it in the marker list and either choose **Edit > Delete Marker** (⌘⌘) from the menu bar, click the trash button on the tool bar, or right-click the marker and choose **Delete** from the contextual menu.

Important: deleting a maker or a marker panel deletes all genotypes at the marker(s).

Copying marker panels and markers

A marker panel can be used with all samples analysed at the same set of markers. You may however use the same marker(s) in different multiplexes (i.e., combined with different markers). If several multiplexes you use share molecular markers, copying panels or markers avoids defining the same marker several times.

To **copy a marker panel or a folder** to the paste board, select it from the [panel list](#) and do Edit > Copy (⌘C) from the menu bar. You may also right-click or ctrl-click the panel/folder and choose **Copy** from the contextual menu. The panel/folder may then be **pasted** to another folder or the same folder by selecting Edit > Paste (⌘V).

Tip: a [description of the panel](#) can also be pasted to a text editor or a spreadsheet application.

To **copy markers between panels**, first select the source panel and markers to be copied in the [marker library](#) and then, either:

- ▶ Choose Edit > Copy from the menu bar, or right-click the markers and choose Copy from the contextual menu. Then select the destination panel and choose Edit > Paste.
- ▶ Drag the rows corresponding to markers and drop them onto the destination panel.

If you try to copy markers that overlap, or have the same name as, existing markers of the destination panel, an alert tells you that these markers could not be copied.

Note: copied markers are independent: changes made to copied markers are not applied to the original markers, and reciprocally.

Applying a marker panel to samples

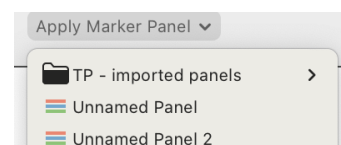
For genotyping, STRyper has to know the microsatellite markers that were amplified for every sample. For this, the correct panel has to be **“applied” to the samples**. A marker panel can be applied to samples before or after [molecular markers](#) are added to the panel.

To **apply a marker panel to samples**, do one of the following:

- ▶ Drag the panel from the [panel list](#) onto the sample table. This applies the panel to all listed samples.
- ▶ Select a panel from the list, click the **Apply To** button below the list and select the samples from the contextual menu (either all listed samples or selected samples).

Tip: if the table lists chromatograms obtained from different marker panels, you may only show those that were amplified with given panel by [filtering](#) (for instance by plate name) .

- ▶ Right-click samples from the [sample table](#) and choose the panel from the **Apply Marker Panel** contextual menu. This applies the panel to clicked/selected samples.
- ▶ Select samples from the [sample table](#) and select a marker panel from the **Apply Marker Panel** button that is visible above traces in the [viewer](#). Note: if the sample(s) displayed have no [size standard](#) or a [sizing quality](#) of 0%. Instead, a text indicates “Cannot apply marker panel” instead.

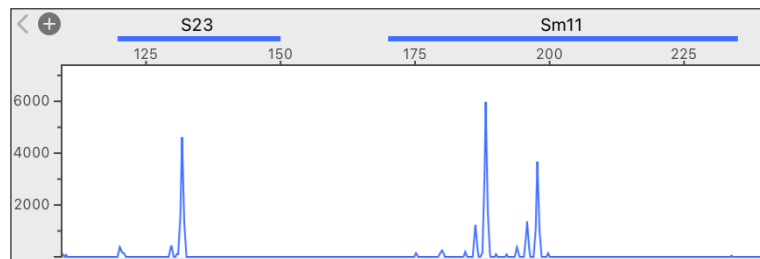


Applying a marker panel has two consequences on each sample:

- 1) A **genotype is initialized** for each marker, which prepares genotype calling. Adding a marker to a panel also initializes a genotype at that marker for every sample already associated with the panel.

Important: applying a marker panel to a sample that was already associated with a panel deletes any genotype that the sample previously had.

- 2) The **viewer** displays the marker(s) above the trace of the corresponding channel. The marker range is presented as a horizontal segment colored according to the marker's dye, below the marker's name.



Markers are not displayed in the viewer if:

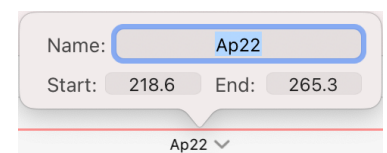
- traces from several channels are **stacked on the same row** (unless the viewer **displays genotypes**).
- samples associated with different marker panels are stacked in the same row. In this case, "Multiple marker panels" is displayed instead of markers.
- the sample sizing quality is 0% or the sample has no size standard applied. In this case, "Cannot show markers" is displayed instead of markers.

Modifying molecular markers

You can rename a molecular marker or change its range at any time, if for example you find new alleles beyond its current range.

To change the range and name of a marker do one of the following:

- ▶ edit (click) the cell of the **Name**, **Start** or **End** column of the **marker library**.
- ▶ double-click a marker label (rectangle) above a trace and enter new values in the popover.
- ▶ click the marker's name or right-click the marker, and choose **Edit Name and Range** from the drop-down menu, which spawns the popover.



The value you enter for the start/end of a marker may be modified to avoid overlap with other markers of the same channel, and to ensure that the marker is at least 2-bp wide and spans its set of **bins**.

To resize a marker and therefore change its range, click and drag its edge.



To **navigate between markers** of the same channel in the [viewer](#), you can:

- ▶ click a navigation button, < or >, above the [top ruler](#),
- ▶ place the cursor over the top ruler or the area showing markers and scroll left or right. This behavior can be disabled by unchecking **Traces > Allow Swipe between Markers** from the menu bar.
- ▶ perform a three-finger swipe on a trackpad, if this gesture is enabled in the trackpad preferences of macOS.

To **zoom to the range of a marker**, click the marker's name above the horizontal ruler, or right-click the marker label, and choose **Zoom to Marker** from the contextual menu.

Defining bins

The positions of peaks representing a given allele slightly vary between runs and samples, due to variations in electrophoresis conditions. STRyper has to know the possible range of these peaks to attribute them to the allele during [genotype calling](#). This range is called **a bin**. Like a marker, a bin has **start** and **end** coordinates in base pairs, and a **name** corresponding to the name of the allele. During genotype calling, any allele detected within a bin is named after it. The allele name appears in the **Allele1** or **Allele2** column of the [genotype table](#). Bins are not strictly necessary for genotype calling. However, without bins, alleles are not automatically named and you may not easily identify peaks corresponding to the same allele.

A microsatellite marker may include dozens of bins, which are separated from each other by the length of the repeat motive, as bins can correspond to all possible alleles of a marker. STRyper allows you to create a set of bins in one operation (see below), [or each bin individually](#).

To **generate bins for a marker**, first do one of the following:

- ▶ If the marker's panel is [applied to samples](#), select at least one of them in the [sample table](#). In the viewer, scroll the trace to the range of the marker. If several samples are selected, you may [stack them in the same row](#).
- ▶ Select a marker in the [marker library](#). The [viewer](#) shows the whole marker's range. **However**, as peaks are not displayed, this visualization does not suggest where to [place bins appropriately](#).

Then, you can either:

- ▶ click the marker's name above traces, or right-click the marker, and choose **Generate Bins** from the contextual menu.
- ▶ Right-click or ctrl-click the view showing traces within the range of the marker and choose **Generate Bins** from the contextual menu.

Ap22 ▾

220 230 240 250 260

Generate bins

Starting at: 220 bp Ending at: 264 bp

Every 2 bp Bin width: 1.0 bp

☒ Remove all existing bins

Cancel Add Bins

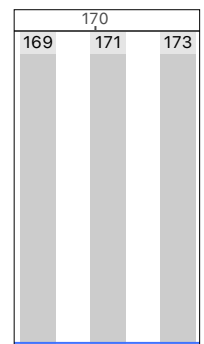
Upon clicking the **Add Bins** button, bins are added at the specified positions, which correspond to their middle. In the example shown above, the first bin starts at 231.5 bp and ends at 232.5 bp.

Tip: you may alternatively [import bins from a text file](#).

Tip: leaving a margin of several base pairs (bp) between the edges of the marker and the first and last bins helps [adjusting bin positions](#) afterwards.

Generating bins removes any existing bin that overlaps with new bins, or all existing bins if you ticked the **Remove all existing bins** checkbox.

The [viewer](#) displays bins as vertical rectangles. Bins are initially named after the middle of their range in base pairs. Some bin names are automatically hidden to avoid overlap. You may need to [zoom in](#) to see all bin names.



Upon generating bins, the viewer enters a mode allowing you to [move bins](#).

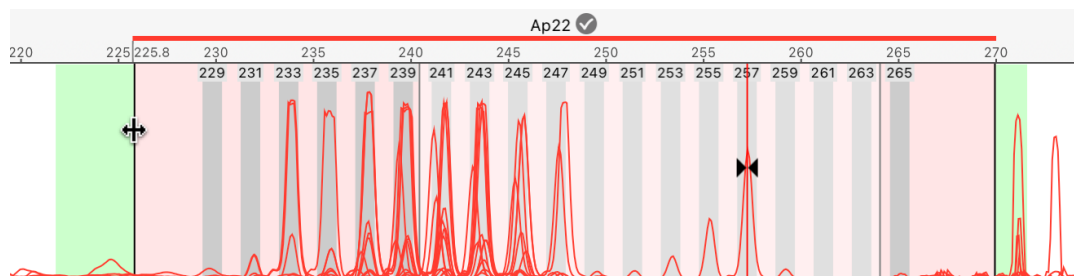
Tip: if a marker contains very few alleles, you may [add individual bins](#) rather than using the above procedure. Depending on your analyses (e.g., population genetics) you need not constrain bins to known alleles, as bins can help to identify new alleles.

Adjusting bin positions

The positions of new bins rarely coincide well with the marker's alleles. This is because microsatellite amplicons do not migrate exactly according to their sizes, as their motility slightly differs from molecular ladder fragments. Here is how to make bin and allele positions match.

First, select several samples showing different alleles, preferably from the same sequencer run or plate, [stack them](#) in the viewer and display the marker range.

To move bins, click the marker's name above the trace, right-click the marker, or right-click in the area within the marker range, then choose **Move all Bins** from the contextual menu. The marker range takes a light pink color. The mouse cursor becomes an open hand over this area to signify that bins can be moved. This affects all rows on which the marker is visible.



To make bins coincide with peaks, move the bin set by dragging the pink rectangle left or right. [Tip: peaks resulting from *stuttering* indicate the probable location of alleles, hence where bins should be.] While you move the bin set, a green area appears underneath to represent the allowed range of the bin set, which ensures that bins remain in the marker's range.

Because the spacings between peaks and between bins often slightly differ, you may need to **shrink or expand the set of bins**. You do so by dragging the edge of the pink rectangle. To better control bin positions during the procedure, place **the anchor** appropriately. **The anchor** is the red vertical line with two converging black triangles on the above image (positioned over bin "257"). This line defines a position that does not move when the bin set is resized (in other words, the bin set is resized "around" the anchor). The anchor is positioned wherever you click in the pink area. You may place the anchor at the middle of a bin that coincides well with a peak or a set of overlapping peaks.

While the bin set is resized, a green area denotes the maximal width that the bin set can take while maintaining bins in the marker's range. Similarly, a translucent light rectangle denotes the minimum width that the bin set can take while avoiding overlap between bins.

To exit the bin adjustment mode, you can either: hit the escape key, click the checkbox ☒ next to the marker's name or click anywhere on the trace outside the pink area. Changes to bins are recorded after each individual adjustment, not upon exiting the bin adjustment mode.

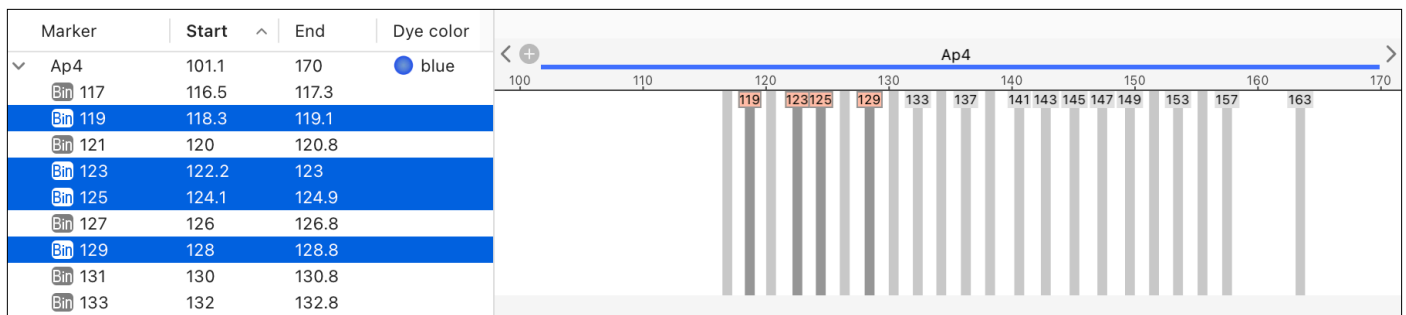
Note: a bin name need not exactly correspond to its position. For instance, a bin named "252" may end up close to 251 bp after adjusting its position. This has no consequences on genotype calling or downward analyzes, which do not rely on absolute allele sizes.

Tip: if certain alleles have odd sizes (due to mutations that do not correspond to indels of microsatellite motives), do not try to make them coincide with bins using this method. You may instead [add/edit individual bins](#).

Viewing and editing bins in the marker library

Bins are listed in the [marker library](#). You can reveal/hide them by clicking the disclosure rectangle next to a marker's name. Each bin has its name preceded by a badge: **Bin**.

Upon selecting markers or bins on the library, bins appear in the viewer. Selected bins are highlighted.



To rename a bin or modify its range, simply edit the corresponding cell of the table. The value you enter may be modified to avoid overlap with other bins, to ensure that the bin is at least 0.1-bp wide and remains within the range of the marker.

To delete bins, select them and either choose **Edit > Delete Bins** (⌘⌘) from the menu bar, click the trash icon  on the [toolbar](#), or right-click them and choose **Delete** from the contextual menu.

Note: you cannot add individual bins directly to the marker library like you can with markers, but you can [import bin sets from text files](#).

Editing bins with the viewer

The [viewer](#) allows adding, removing and modifying bins individually.

To add/edit individual bins, first do one of the following:

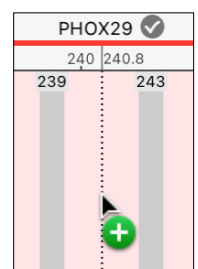
- ▶ right-click the marker above traces or click its name, and choose **Edit Bins**.
- ▶ Right-click or ctrl-click the viewer in the range of a marker and choose **Edit Bins Manually** in the contextual menu.
- ▶ If a bin is already present and if you use a trackpad with force touch enabled, perform a force click on the bin. This activates the bin editing mode and selects the clicked bin.

The marker's range takes a light pink color. The mouse cursor gets a green '+' sign over this area to signify that bins can be added. In this editing mode, genotypes at the marker [cannot be edited](#), and [saturated regions](#) and [peak information](#) cannot be displayed in the marker's range. This affects all rows in which the marker is visible.

To add a bin, you may either:

- ▶ click and drag on the pink area (outside existing bins) to define the range of the new bin.
- ▶ double click in the pink area. A new one-bp-wide bin (or less depending on the available space) is created at the mouse location. No bin is added if there is less than 0.2 bp between existing bins.

These actions spawn a popover allowing to specify a name and range for the bin.



To change the name and range of a bin, double-click it. This spawns a popover allowing to specify a name and range.

To resize a bin, click and drag one of its edges. You may also **move a bin** by dragging it.

To exit bin editing, you can either: hit the escape key, click the checkbox  next to the marker's name or click anywhere on the trace outside the pink area.

Hiding bins

You can **hide bins** to make peaks more visible. To do so, click the [Show/hide bins button](#)  or deselect **Traces > Bins** (^B) from the menu bar.

Note: bins always stay visible while they are being edited or moved.

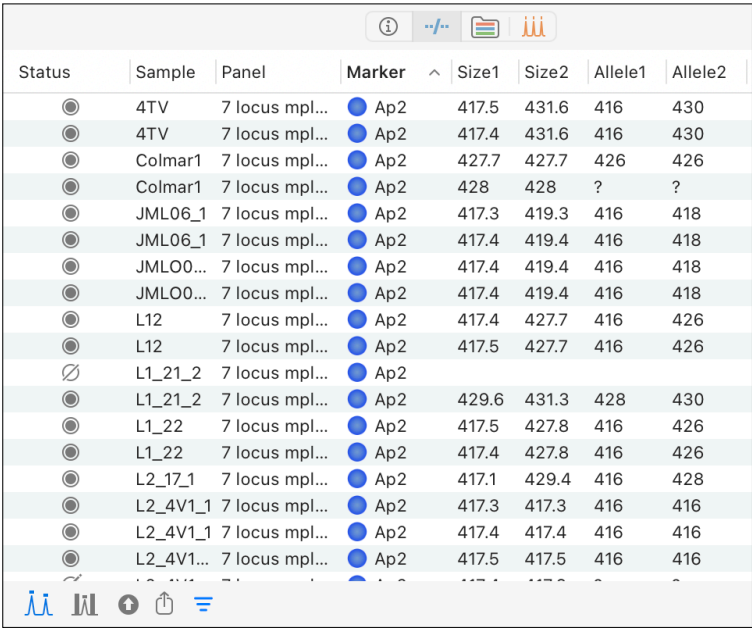
Genotyping

In STRyper, a **genotype** corresponds to the allele(s) of a given sample at one molecular marker. When you [apply a panel of molecular markers to samples](#), a genotype is initialized for each marker and sample. STRyper is now ready for genotype calling.

The genotype table

[The genotype table](#) lists the genotypes of the samples that appear in the [sample table](#). To show the **genotype table**, click the second button of the segmented control  or choose View > Genotype Table (⌘⌘G) from the menu bar.

As each row represents one genotype (hence one marker and sample), the table can list the genotypes of samples analyzed with different marker panels.



Status	Sample	Panel	Marker	Size1	Size2	Allele1	Allele2
●	4TV	7 locus mpl...	Ap2	417.5	431.6	416	430
●	4TV	7 locus mpl...	Ap2	417.4	431.6	416	430
●	Colmar1	7 locus mpl...	Ap2	427.7	427.7	426	426
●	Colmar1	7 locus mpl...	Ap2	428	428	?	?
●	JML06_1	7 locus mpl...	Ap2	417.3	419.3	416	418
●	JML06_1	7 locus mpl...	Ap2	417.4	419.4	416	418
●	JMLO0...	7 locus mpl...	Ap2	417.4	419.4	416	418
●	JMLO0...	7 locus mpl...	Ap2	417.4	419.4	416	418
●	L12	7 locus mpl...	Ap2	417.4	427.7	416	426
●	L12	7 locus mpl...	Ap2	417.5	427.7	416	426
○	L1_21_2	7 locus mpl...	Ap2				
●	L1_21_2	7 locus mpl...	Ap2	429.6	431.3	428	430
●	L1_22	7 locus mpl...	Ap2	417.5	427.8	416	426
●	L1_22	7 locus mpl...	Ap2	417.4	427.8	416	426
●	L2_17_1	7 locus mpl...	Ap2	417.1	429.4	416	428
●	L2_4V1_1	7 locus mpl...	Ap2	417.3	417.3	416	416
●	L2_4V1_1	7 locus mpl...	Ap2	417.4	417.4	416	416
●	L2_4V1...	7 locus mpl...	Ap2	417.5	417.5	416	416

The columns show:

- the [status](#) of the genotype,
- the **names** of the genotyped **sample**, of the **marker**, and of the **panel** to which the marker belongs,
- the **size** of each allele (**Size1** and **Size2** columns) as estimated via the molecular ladder, based on the [scan](#) at the tip of the peak. This size is not the true allele (amplicon) size, which is an integer. **Size1** is the size of the shorter allele.
- the **name** of each allele in the **Allele1** and **Allele2** columns, **Allele1** being the name of the shorter allele,

- the **height** of the peak induced by each allele (in RFU, [without baseline level removal](#)). The Height1 column shows the height of the peak induced by the shorter allele. These columns are hidden by default.
- **notes** that can be added to the genotype by typing in the corresponding cell,
- a description of [additional peaks](#) found during genotype calling,
- the [marker offset](#) for the genotype.

Before [genotype calling](#), alleles have no size and no height as they are not assigned to peaks yet. They have the name attributed to missing alleles (by default, an empty string).

You can reorder, hide columns, and sort the genotype table like you can for the [sample table](#).

Tip: rows of the genotype table can be [exported to text file](#) or copied to the pasteboard.

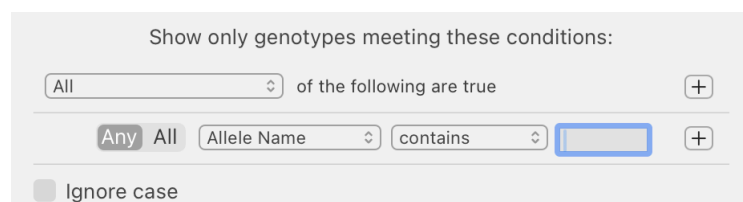
To identify the samples associated with genotypes, first select genotypes in the genotype table, and either:

- ▶ press the arrow button  below the table.
- ▶ right-clicking or ctrl-click the selecting genotypes and choose **Show Source Sample(s)** from the contextual menu.

These actions select the corresponding samples in the sample table.

To identify the genotypes of specific samples, right-click or ctrl-click these samples in the [sample table](#) and choose **Select Genotype(s)** from the contextual menu. This action selects the corresponding rows of the genotype table. If these samples have no panel applied or if all their genotypes are [filtered out](#), the menu item is disabled.

To filter the genotype table and display genotypes meeting your criteria, press the “filter” button  below the table. A popover allows you to define filter criteria. You may filter according to the name, the size or the intensity (height) of alleles. If both alleles of a diploid genotype have to meet the condition you define, choose the **All** on the segmented button. If only one of the alleles has to meet the condition, choose **Any**. This button has no effect on genotypes at haploid markers.



A filter on genotypes is associated to a [folder](#) and is recorded between app launches. If the selected folder has a filter applied to it, the filter button takes this appearance: .

Note: if genotypes are modified so that they meet or no longer meet filter criteria, you have to reapply the filter or select another folder and go back to the previous folder, to update the genotype table.

To remove a filter on genotypes, press the filter button  below the genotype table and click the Remove Filter button in the popover.

Tip: you can maintain a list of all genotypes meeting certain criteria. For this, create a [smart folder](#) that contains all samples of the database (for instance, by setting an import date that is more recent than 2020) and apply a filter to the genotypes on that folder.

Genotype calling

Genotype calling consists in (i) detecting peaks that represent alleles, which determines allele sizes, and (ii) attributing names to alleles if the marker has bins. Each allele is named after the bin that spans the tip of its peak.

The **status** of genotype calling is represented by an icon in the **status** column of the [genotype table](#):

- : The genotype has not been called nor [edited](#). Alleles have no size and have the name given to missing alleles, which is blank by default.
- : The genotype has been called and alleles were found.
- : No allele was found during genotype calling.
- : The genotype has been [edited manually](#).
- : The marker has been modified (range, bins) since genotype calling/editing. This status denotes that genotypes be be called again to account for new bin names, positions, etc.
- : Sample sizing ([marker offset](#), molecular ladder, [fitting method](#)) has been modified since genotype calling. Allele estimated sizes may have changed.
- : The sample [sizing quality is 0](#) or no size standard has been applied to the sample. In this case, the genotype cannot be called nor edited, alleles have no size and no name.

Note: genotype status does not represent the reliability of genotype calling.

To call genotypes, do one of the following:

- ▶ Click the Find Alleles button  below [the genotype table](#). This action calls genotypes, or re-calls genotypes that have not been edited manually (whose status is not .
- ▶ Select and right-click genotypes from the genotype table and choose Find Alleles from the contextual menu. This menu is not available if the samples don't have proper sizing.
- ▶ Select and right-click samples from [the sample table](#) and choose Find Alleles from the contextual menu (these samples have to be properly sized and have a panel containing at least one marker applied to them). This action calls genotypes for markers of the panel.

After genotype calling, alleles outside existing bins take the name attributed to out-of-bin alleles (by default, "?"). If a marker has no bin, alleles get no names. If no suitable peak was found in the marker range, alleles get the name attributed to missing alleles, by default a blank string.

To change the default names for non-binned or missing alleles, display the application [settings](#) and enter the name of your choice in the appropriate field. This change does not affect genotypes that have already been called.

Allele binning

Allele binning names alleles after the bins that overlap the tips of peaks that were already identified, and therefore perform only the second task of genotype calling. You may need to (re-)bin alleles that were previously found if, for instance, you have modified/moved bins or edited the molecular ladder (status is ● or ⚠). Binning does not change your [manual assignments](#) of peaks to alleles.

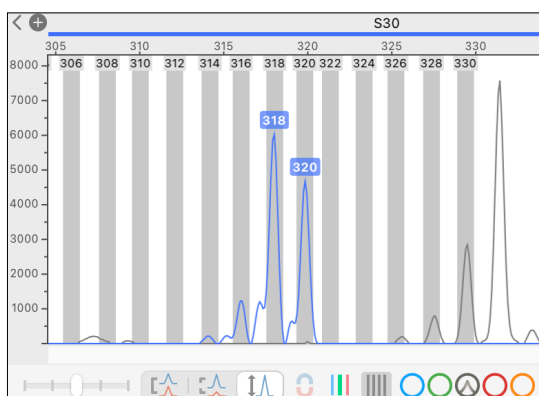
To bin alleles, you can either:

- ▶ Click the Bin Alleles button  below [the genotype table](#). This action ignores genotypes that have been edited manually (status ⚙).
- ▶ Select and right-click genotypes from [the genotype table](#) and choose Bin Alleles from the contextual menu.

For genotypes that have not been called (status ○), these actions perform genotype calling.

Displaying alleles in the viewer

To display alleles in the viewer, select genotypes from [the genotype table](#). If necessary, click the left segment of the [stack trace mode selector](#): . The [viewer](#) displays the corresponding traces in the marker's range, as shown below.



Alleles names appears above peaks as labels whose color corresponds to the dye of the marker. If an allele has no name (because the marker has no bin or you deleted its name), the size of the corresponding fragment in base pairs appears instead.

Note: Allele labels are also visible when [viewing samples](#), unless several samples or channels are [stacked on the same row](#).

Each row of the viewer corresponds to one selected row of the genotype table (in the same order) and you cannot stack several samples per row. Horizontal positions cannot be [synchronized between rows](#). Each row can be scrolled horizontally and zoomed, but its visible range is reset to that of the marker each time the genotype is selected in the genotype table.

You may choose to [display several channels](#) (colors) per row. On the image above, the black channel is displayed for a genotype of a marker labelled with the blue dye. The blue curve thus appears even though the blue button is not ticked. Displaying several channels may be help to determine if peaks are subject to [crosstalk between channels](#).

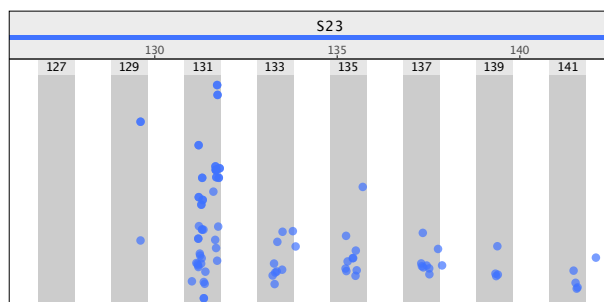
Displaying alleles as dots

STRyper can represent alleles as dots in the [viewer](#). In this visualization mode, all alleles of the same marker appear on the same row. This mode helps comparing allele sizes and spotting rare/unusual alleles.

To visualize alleles of the same marker as dots on a single row, first select genotypes in the [genotype table](#), then:

- ▶ click the right-hand segment of the [stack trace mode selector](#):  or
- ▶ choose **Traces > Stack Genotypes per Marker** from the menu bar.

Alleles appears as colored dots whose vertical positions represent the heights of corresponding peaks. No trace is displayed.



To highlight the genotype corresponding to an allele, click the dot representing the allele. The row corresponding to the genotype in [the genotype table](#) gets highlighted with a white frame.

To display the chromatogram associated with an allele, right-click or ctrl-click the dot representing the allele and choose View Chromatogram from the contextual menu. This selects the corresponding genotype and the chromatogram appears in the viewer.

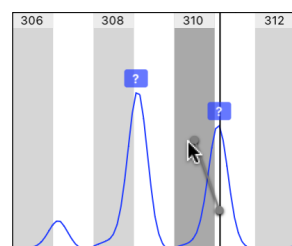
Tip: visualising alleles as dots can help to adjust bin positions. You may [move the set of bin](#) to make bins coincide with clusters of dots, each of which is assumed to represent the same allele. Once bin positions are adjusted, perform [allele binning](#). **Important:** make sure that the peaks representing alleles are not clipped due to [saturation](#), in which case the position of peak tips (hence of alleles) is not accurate.

Manual genotype editing

You may need to **attribute alleles to peaks manually** in case of errors in genotype calling. You can even forgo genotype calling, as STRyper allows manual assignment of peaks to alleles using the [viewer](#).

To assign a peak to an allele in the marker range, you can either:

- ▶ Double-click the peak, or right-click it and choose **Add Allele** from the contextual menu. If the peak has no allele, it gets one (possibly using an allele assigned to another peak, to maintain the ploidy to the marker). If the peak already has an allele, double-clicking it removes the allele, which may move to another peak if that other peak already has an allele (thereby creating a homozygote).
- ▶ Click the peak and drag the mouse. If the marker has [bins](#), a handle starting at the horizontal position of peak's tip appears. Drag the other end of the handle to the bin and release the mouse button. The peak takes an allele named after the bin. If you release the mouse outside a bin, the allele gets the "out of bin" name (by default, "?") unless you release the handle above or below the view showing the trace, in which case the action has no effect.
- ▶ Drag an allele label from another peak to the desired peak. The allele gets named after the bin present at the peak's tip, or the "out of bin" name if there is no such bin.



Note: you cannot assign alleles to peaks that are outside the marker range. If such peak should be considered as an allele, you have to [expand the marker range](#).

If a peak that should be an allele was not detected if its fluorescence level is less than 100 units, you may [insert a peak manually](#).

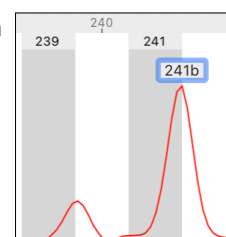
To remove an allele (without creating a homozygote) you can either:

- ▶ select the allele label and choose **Edit > Delete Allele** (⌘) from the menu bar, or click the trash icon  on the [toolbar](#).
- ▶ drag the allele label above the top edge of the row showing traces.

This removes the allele's size and gives it the name of missing alleles (by default, an empty string).

To give an allele a custom name, do one of the following:

- ▶ Enter the name in the Allele1 or Allele2 column of the [genotype table](#).



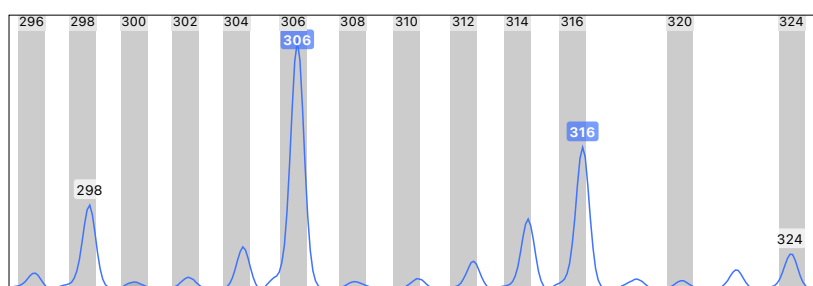
- ▶ Double-click the allele label above the peak and type the name.

If you delete an allele name, its size in base pairs appears above its peak instead.

Tip: to quickly review genotypes, [sort](#) the genotype table by marker, allele1 and allele2 in ascending order (assuming allele names correspond to sizes). Then, view genotypes from top to bottom. This method helps detecting errors in allele call.

Additional peaks

In addition to alleles, STRyper may annotate other peaks found in a marker range if they meet certain criteria. These peaks are labelled like alleles, except that their names appear in black over a white background. Additional peaks can help to identify contamination between samples, insufficient specificity of the PCR or polyploidy.



Additional peaks can be [edited](#) like alleles: their labels can be moved between peaks and renamed by double-clicking, and they can be deleted by the **delete** key. In addition, they can be deleted by double-clicking their peak. These actions have no effect on the sample's genotype as defined by the **Allele1** and **Allele2** columns of the genotype table.

If you consider an additional peak as a real allele, drag an allele label onto the peak or right-click the peak and choose **Add Allele** from the contextual menu. These actions remove the additional peak label.

To label a peak as an additional peak right-click the peak and **Add Label** from the contextual menu. This menu is not available if the peak has an allele. The allele label has to be removed first.

To remove an additional peak, you can either:

- ▶ select the peak label and either hit the delete key, choose **Edit > Delete Label** (⌘) from the menu bar, or click the trash icon  on the [toolbar](#).
- ▶ drag the peak label above the top edge of the row showing traces,
- ▶ double-click the peak.

To remove all additional peaks from several genotypes, select these genotypes in the [genotype table](#), right-click them and choose **Remove Additional Peaks** from the contextual menu.

Information on additional peaks appear in the **Additional Peaks** column of the genotype table. For each peak, the column shows the size (in base pairs, rounded

Additional Peaks
248.7:248 264.0:264

to the nearest decimal) of the additional peak and its name, separated by a colon. Data from different peaks are separated by spaces and are sorted by increasing size. This column is not editable.

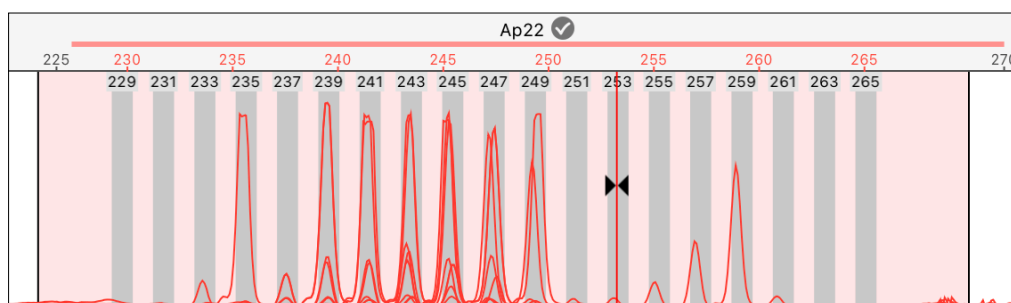
To disable the annotation of additional peaks during genotype calling, untick the corresponding checkbox in the [settings window](#).

Marker offsets

Electrophoresis conditions (temperature, instrument, gel...) affect amplicons and DNA ladder fragment differently, which results in variations in the estimated sizes for the same allele. These variations may move alleles out of their respective [bins](#). Widening bins is not a good solution to this problem because wide bins (>1bp) limit the capability to discriminate alleles and to identify mutant alleles.

To address the issue, certain applications allow defining several bin sets per marker. This requires that any modification on one bin set (such as adding bins for new alleles) be replicated in other bin sets of the marker, and to use different bin sets depending on sequencing conditions. STRyper has a better solution to this issue. This solution consists in moving bins so that their positions coincide with alleles of the samples you choose, *without modifying bins*.

To adjust bin positions for specific samples, select the samples in the [sample table](#) and [stack them in the same row](#). Then choose **Adjust Offset** from the marker drop-down menu (accessed via right-clicking the marker or by clicking its name). Then, [move bins](#) such that their positions match those of peaks. **Important:** to associate peaks with the correct bins, you must be able to determine the peak(s) corresponding to at least one known allele. Ideally, at least one chromatogram should represent a sample of known genotype.



To exit this editing mode, either hit the escape key, click the checkbox ☒ next to the marker's name, click anywhere on the trace outside the pink area, or select other samples.

This operation does not modify bin coordinates. Indeed, sizes of the [top ruler](#) move in sync with bins. This operation defines **a marker offset**, which changes **the estimated size of alleles within the marker range**. This offset applies a linear combination of the form $y = a + bx$, where x is the size of a peak

computed thanks to the molecular ladder, and y is the size used in the marker's range. Parameters a and b are computed from the shift you applied to the bins.

A chromatogram with an offset at a marker shows red graduations on the top ruler in the range of the marker, and the bins appear shifted compared to samples without the offset.

Note: if samples with different offsets for the same marker are [stacked on the same row](#), the offset of bins and of sizes on the top ruler corresponds to a random sample shown at the row.

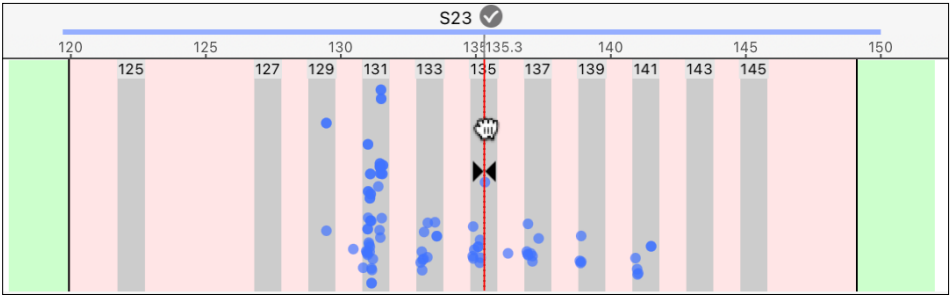
Peaks that are outside a marker range, as represented by the colored segment above traces, cannot be assigned to alleles even if they are within the pink area representing the marker range with its offset.

Important: adjusting a marker offset can visually move bins out of the marker's range. Bins from markers that are very close to each other might thus overlap. This situation is not managed by STRyper and must be avoided. If you use marker offsets, leave sufficient space between bins of different markers of the same color.

Because a marker offset is specific to a sample and a marker, it is an attribute of the corresponding *genotype*. This offset appears in the **Offset** column of the [genotype table](#). The numbers in parentheses are terms a and b of the offset.

Offset
(6.2, 0.978)
(6.2, 0.978)
(6.2, 0.978)
(6.2, 0.978)

You can also **adjust a marker offset when alleles are [displayed as dots](#)**. The procedure is the same, except that adjusting the offset changes the position of alleles (dots) rather than bins and graduations of the top ruler.



The result is true to the nature of a marker offset: a correction applied to *allele sizes* rather than bin positions. (When chromatograms are displayed, the offset affects bins rather than alleles because alleles may not be displayed, if for instance several traces are stacked or genotypes have not been called.)

Important: make sure that the peaks representing alleles are not clipped due to [saturation](#), in which case the positions of peak tips (and dots) do not represent allele sizes accurately.

To remove offsets applied to genotypes, select genotypes in the [genotype table](#), right-click them and choose **Remove Offset(s)** from the contextual menu. You may alternatively select one or several samples in the [sample table](#), click the name of the marker label above traces, or right-click it, and choose **Remove Offset** from the drop-down menu.

When to adjust marker offsets

Adjusting a marker offset has the same purpose as [adjusting bin position](#): to make alleles coincide with bins. You may therefore wonder which procedure to apply.

You should **adjust bin positions** for the first batches (e.g., sequencing plates) of samples you analyze after defining a new set of bin, and under uniform sequencing conditions (same instrument, gel, etc.). If electrophoresis conditions are stable, alleles should remain in their respective bins in future sequencing runs.

You should **adjust the offset of a marker** for samples analyzed in subsequent runs/plates if their alleles appear shifted in their position compared to bins.

Copying and pasting marker offsets

Via copy & paste, you can apply an offset defined for some reference sample(s) to other samples sequenced in similar conditions.

To copy the offset of one marker to the pasteboard, you can either:

- ▶ Select the sample(s) or genotype(s) and [display the marker](#) in the viewer. Then click the marker's name and choose **Copy Offset** from the contextual menu. This action is not available if the marker has no offset, or if alleles are displayed as dots and represent genotypes with different offsets.
- ▶ Select one genotype from the [genotype table](#) and choose **Edit > Copy (⌘C)** from the menu bar, or right-click the genotype in the table and choose **Copy** from the contextual menu.

To copy the offsets of all the markers analyzed on a sample to the pasteboard, select the sample in the [sample table](#) and either:

- ▶ choose **Edit > Copy** from the menu bar,
- ▶ right-click the sample and choose **Copy** from the contextual menu.

Note: if several samples or genotypes are selected, no marker offset is copied.

To apply a copied marker offset to other samples/genotypes analyzed at this maker, select these samples/genotypes from the [sample table](#) or [genotype table](#) respectively. Then either:

- ▶ Right-click or ctrl-click the sample(s) or genotype(s) and choose **Paste Offset(s)** from the contextual menu. If you copy & paste marker offsets between samples (not genotypes), you have to choose the marker(s) to which you paste the offset, via a submenu.
- ▶ [Display a marker](#) in the viewer, click the marker's name and choose **Paste Offset** from the drop-down menu. This action applies the offset to the genotype(s) or sample(s) displayed at the row. You may therefore [stack samples in the same row](#) or [display alleles as dots](#) to apply an offset to several samples/genotypes.

Export/import

Exporting genotypes

To **export the [genotype table](#)** as a tab-delimited text file, Go the genotype table and click the [toolbar](#) Export button, which should look like this: . If it does not, click anywhere in the genotype table. Alternatively, click the export button  below the genotype table. An export window lets you choose to export the whole table or only the selected rows, if any.

You can also **export certain genotypes only**. For this Right-click or ctrl-click selected rows from the [genotype table](#) and choose Export to Text File... from the contextual menu.

To **add sample metadata to the exported genotypes**, tick the Add Sample-related Columns checkbox in the export panel. Each visible field in the [sample table](#) will be added to each genotype.

Tip: [change the column visibility](#) to choose the columns to be exported.

To **copy rows of the genotype table** to the pasteboard, select rows and choose Edit > Copy ( C), from the menu bar. The rows can then be pasted to a spreadsheet or text editor.

Exporting marker panels

You can **export marker panels as a text file** for archiving, editing markers/bins with a text editor, or transferring the panel descriptions to colleagues. This text file describes all markers and bins of the panel(s). You can export a single panel or all panels contained in a folder. For this, do one of the following:

- ▶ select the panel/folder from the [panel list](#) and click the Export button  on the [toolbar](#).
- ▶ select the panel/folder and choose File > Export Panel(s) to file...(⌘E)
- ▶ right-click the panel/folder and choose Export to Text File... from the contextual menu.

These actions are not available if the selected folder does not contain any panel.

The exported file is a UTF-16 text file formatted as this example:

```
panel    some_panel
# an optional comment line that can be added anywhere
marker  marker1    154.92    180.73    red    2    4
bin     167    166.75    167.40
bin     164    163.57    164.45
bin     165    164.70    165.44
bin     163    162.60    163.32
bin     166    165.60    166.46
bin     168    167.64    168.63
```

```

bin      162    161.55    162.35
bin      161    160.54    161.31
bin      176    176.03    176.91
marker   marker2      242    275.65      red    2      3
bin      253    252.31    253.03
bin      255    254.14    255.05
bin      270    269.44    270.42
panel    new_panel 2
marker   marker2      155    180    black 2      4
bin      167    166.75    167.40
bin      164    163.57    164.45

```

- Fields are delimited by **tabulations**.
- Lines starting with **#** can be used to add comments.
- The first uncommented line starts with the word **panel** followed by the panel name.
- The next line defines the first marker. The line starts with the word **marker**. The following fields represent in this order: the marker name, the start coordinate, the end coordinate, the channel color (either **blue**, **green**, **red**, **black** or **orange**), the ploidy (either **1** or **2**) and the motive length (2 to 7).
- The subsequent lines define the marker bins. Each row must start with the word **bin**. The following fields specify the bin name (which need not be a number), start and end coordinates. Bins of a marker, and markers of a panel, can be described in any order, but all bins of a marker must be described before the next marker.
- Decimal separators are **periods**.

Note: STRyper allows most Unicode characters (excluding tabs) in panel, marker and bin names. However, for compatibility with other applications, roman characters are recommended.

Tip: when a [panel is copied](#), a text following the above format is copied to the pasteboard and can and pasted to a text editor.

Importing marker panels

STRyper can **import marker panels from text files** conforming to the above [specifications](#), as well as panel files exported by GeneMapper (version 4). Here is an example of such file:

```

# This a Genemapper v4 panel file
Version GM v 4.0(not checked)
Kit type:      MICROSATELLITE(ignored)

Chemistry Kit (Ignored)  none
Panel  new_panel  none
marker1 Red  154.92    180.73    -    4    0    none  none  false
marker2 Red  242    275.65    -    3    0    none  none  false
Panel  new_panel 2
marker3 Yellow 155  180    -    4    0    none  none  false

```


You may refer to the GeneMapper user guide for a description of this format. Elements in grey can be omitted.

Important: files must end with the extension ".txt", and the word **Version** must appears in the first field before the first panel of a GeneMapper file is described, as this keyword is used to determine the file format.

Markers imported from a GeneMapper file are considered **diploid** and do not contain bins. You have to [import bins](#) separately.

To import marker panels from a text file, do one of the following:

- ▶ Click the Import markers button  on the [toolbar](#) or choose File/Import Marker Panel(s)... from the menu bar. The imported panel(s) appear(s) at the bottom of the [panel list](#).
- ▶ Click the + button at the bottom of the panel list and choose Import Panels from File in the contextual menu.
- ▶ Right-click a folder in the [panel list](#) and choose Import Panels from File... from the contextual menu. The panel(s) will be imported into that folder.

If several panels are imported from one file, they are gathered in a new folder named after the file.

The import process ignores blank lines, lines starting with # and leading/trailing spaces from fields. If the imported file does not conform to the specifications, or if markers/bins overlap, are too short, or exceed the allowed range, an error message describes the issue(s).

Importing bins

If you imported markers from a GeneMapper file, these markers contain no bin. You thus need to import bins from another file generate by GeneMapper: a binset description file. The GeneMapper 4 binset format looks as follows:

```
# This is a comment line
Version GM v 4.0 (not checked)
Chemistry Kit (ignored)
BinSet Name (ignored)
Panel Name a_panel
Marker Name marker1
383 363.825 0.5 0.5 dark gray
386 367.65 0.5 0.5 dark gray mutant
392 373.18 0.5 0.5 dark gray
395 375.605 0.5 0.5 dark gray
398 379.095 0.5 0.5 dark gray mutant
401 382.235 0.5 0.5 dark gray
404b 385.395 0.5 0.5 dark gray
Marker Name marker2
193 173.72 0.46 0.47 dark gray
```

196?	176.93	0.43	0.43	dark gray
203	183.19	0.5	0.5	dark gray
207	186.93	0.5	0.5	dark gray
209	188.95	0.5	0.5	dark gray

You may refer to the GeneMapper user guide for a description of this format. Elements written in grey are ignored and can be omitted.

You may also import bin descriptions from a file [exported by STRyper](#). This allows transferring bins between panels containing the same markers if, for instance, you have added new bins to account for newly discovered alleles and want to use this set of bins on another computer. For this, you may [export the panel](#), then import the bins from the exported file into the equivalent panel on the other computer. Importing the file [as a new panel](#) would not replace bins from the existing panel, it would create a new panel that is not associated with any sample. By contrast, importing bins into a marker panel replaces existing bins.

To import bins for markers of a panel do one of the following:

- ▶ select the panel in the [panel list](#) and click the Import Bins button below [marker table](#),
- ▶ right-click or ctrl-click the panel and choose Add Bins from File from the contextual menu,

then choose the file to import.

The import fails if bins of the same marker overlap, if a bin falls outside the range of its marker, or if the file does not refer to the panel or its markers by their exact names.

Importing bins updates the [viewer](#) with the new bins for every sample already analysed at the panel. Every called genotype of these samples gets [status](#) , to denote the fact that bins have changed.

The imported file need not define bins for every marker of the panel. Markers whose names do not appear in the file are not affected by the import.

To import bins for only certain makers of a panel, you have two options:

- ▶ Make sure that all lines related to the other markers of the panel, and their bins, are absent from the file you import.
- ▶ In the file, rename any marker you want to leave untouched, using a name that does not match any marker of the panel. The application skips markers whose names do not match those of markers listed in the [marker library](#).

Exporting and importing folders

A [folder](#), with all its samples and subfolders (recursively) can be exported as a single file and imported afterwards. The exported archive contains all genotypes, size standards and marker panels (including markers and bins) applied to the samples contained in the folder. An archive has the following icon:



To export a sample folder, do one of the following:

- ▶ Select the folder in the [folder list](#) and click the export button  on the [toolbar](#).
- ▶ Select the folder and choose File > Export Folder... (⇧⌘E) from the menu bar.
- ▶ Right-click the folder and choose Export to Archive... in the contextual menu.

If an exported folder is a [smart folder](#), it is exported as a regular folder. Its search criteria are not encoded in the file.

To import a folder from an archive, do one of the following:

- ▶ drag and drop an archived folder into the [folder list](#). You can drop only one archive at a time.
- ▶ click the Import archive button  on the [toolbar](#) and chose the folder to import.
- ▶ choose File > Import Archived Folder... from the menu bar and chose the folder to import.

The last two methods place the imported folder at the bottom of the [folder list](#).

Any marker panel applied to samples present in the archived folder is placed in a new folder named after the imported folder, unless this panel is already present in the database (i.e., has the same name, markers and bins, with the same attributes: name, start, end, etc.). The new folder appears in the [panel list](#).

Any size standard not already present in the database (with the same ladder fragment sizes) is imported as well and the string "-imported" is appended to its name.

Printing chromatograms

You can **print chromatograms** or **export them in a pdf file** as they appear in the viewer, via the Traces > Print Traces... (⌘P) menu.

The printing scale is adjusted to make rows of the viewer fit in the page width, if necessary. Bins, markers, alleles and other labels are printed but always appear deselected. Buttons and scrollers are not printed.

Where is my data stored?

Your data is stored in `~/Library/Application support/STRyper/`. You may access this folder from the Finder by selecting `Go > Go To Folder...` (⌘G) and pasting this path into the field.

Can I change the location of the database?

No, you cannot. STRyper looks for its database in `~/Library/Application support/STRyper/`. Moving the database elsewhere will make the application recreate a blank database in this folder. However, you may replace the database folder with a symbolic link named identically and pointing to another folder.

Is my data backed up?

STRyper saves its database every 30 seconds (if necessary) and after most actions like sample import, but it does make backup copies. We recommend to regularly backup `~/Library/Application support/STRyper/`.

What if STRyper could not load the database?

There is nothing you can do to fix a corrupt database beyond replacing it with a backup. If the app cannot load the database and therefore cannot be used, you may delete or move the folder `~/Library/Application support/STRyper/` to another location and relaunch the application. You may additionally [report the issue](#).

How can I work on the same project on different computers?

You can [export and import folders](#) between computers. Transferring/synchronizing the whole [database](#) should work but has not been tested. If you wish to do so, make sure that the database is fully transferred before opening STRyper and to use the same version of STRyper between computers.

Is there an equivalent to GeneMapper's bin sets?

You may define several panels with different sets of bins for the same markers, but STRyper has a more flexible solution: defining the [offset of a marker](#) for specific samples.

Why don't the size standards listed in the library have dyes names?

The [size standard library](#) lists, for instance, "GeneScan 500" rather than "GeneScan 500 **LIZ**", "**LIZ**" being the dye used. This is because the dye is irrelevant to sizing, only the sizes of fragments matter.

Simply ignore the dye used by the molecular ladder, just make sure to apply the size standard with appropriate sizes.

Why aren't certain size standards included in the library?

Size standards including long fragments (≥ 800 bp) violate assumptions used by STRyper regarding fitting methods. If you analyse fragments that exceed ~ 800 bp (which is not recommended for microsatellites, as size estimation is less accurate for longer sizes), you may need another software.

As of version 1.6, STRyper no longer includes the GenesScan 1000/1200 (these size standards remain if a previous version of STRyper was used). You can still define [custom](#) equivalents to these size standards, but we advice against defining sizes longer than 800 bp.

Can I use allelic ladders?

An allelic ladder, also called "inter-lane standard", contains a mixture of known alleles for one or several markers, to help control bin positions with respect to alleles. STRyper considers a chromatogram file obtained from such mixture as a regular sample. You may however use this sample to efficiently [adjust the offset](#) of a marker for all samples analyzed in the same conditions.

Why was this peak not attributed to an allele?

This may be because the peak is considered by the application as resulting from crosstalk, i.e., it does not represent a DNA fragment from the channel that is visualized, but from another channel. You can check this by showing peak tooltips. You can still [assign an allele](#) manually to the peak.

Alternatively, the peak may be too faint to be detected if its fluorescence level is less than 100. In this case, hovering it with the mouse does not make a vertical line appear at its tip. You can still [force the detection of the peak](#) and assign it to an allele.

Can I use STRyper for forensics or diagnostics?

No. STRyper comes with no guarantee whatsoever and is intended for research use only.

Does STRyper assess genotyping reliability?

No. Genotyping quality must be assessed by the user. In our experience, genotyping quality computed by applications is poorly informative.

Is there a history of changes applied to genotypes?

STRyper does not maintain a history of manual changes applied to genotypes, as this is not important to most researchers. You may still add notes to a genotype in the **Notes** column of the [genotype table](#).

How can I export genotypes with only one row per individual for all markers?

You cannot. This is because samples from the same folder can be analyzed at different marker panels. However, if samples are analyzed at the same markers in the exported genotype file, you can achieve what you want with [R](#):

```
requires(data.table)
genotypes <- fread("genotypes.txt", header = TRUE, sep = "\t") # use the
appropriate path for "genotypes.txt"

# if you wish to paste allele names into genotype strings:
genotypes[,genotype := paste(Allele1, Allele2, sep = "")]
genotypesPerIndividual <- dcast(genotypes, Sample ~ Marker, value.var =
"genotype")

# if not:
genotypesPerIndividual <- dcast(genotypes, Sample ~ Marker, value.var =
c("Allele1", "Allele2")) # you may use c("Allele1", "Allele2", "Size1", "Size2")
if required

# You may include more columns in the formula, e.g.: Sample + Panel + Well ~
Marker, assuming these columns are present in the table
```

Can I analyze polyploid individuals?

STRyper only manages haploid and diploid markers. At a polyploid marker, the genotype cannot directly be determined from the number of peaks. However, you may use [additional peaks](#) to annotate additional alleles. This could be a solution to genotype polyploid individuals.

Can I use STRyper for other markers than microsatellites, such as AFLPs?

You should probably not. STRyper was designed for microsatellite markers.

How can STRyper be used in a multi-user environment?

As for any regular app, the user of STRyper is the user logged in the operating system. If several users have to use the app on the same computer and work on different projects, the computer administrator has to create a user account for each.

Different users can of course separate their projects into different [folders](#), but these folders are accessible by everyone using the app from the same user account.

Where can I report bugs?

Please, report bugs at <https://github.com/jeanlain/STRyper/issues>.

How can I cite STRyper in my article?

Please cite [STRyper: A macOS application for microsatellite genotyping and chromatogram management](#).

Will STRyper be ported to Windows?

The author of STRyper does not have the resources to port the application to other platforms. To make porting possible, direct a Windows developer to <https://github.com/jeanlain/STRyper>.